

Quantum Mechanics

There are only a few problems in quantum mechanics that can be solved exactly, most notably the harmonic oscillator, a particle in a box, and the hydrogen atom. Nearly all other nontrivial quantum problems either have no known analytic solutions or can be attacked analytically only with extreme difficulty. This is why perturbation methods play such an important role in quantum theory and also why numerical methods are an attractive alternative. Indeed, an extremely wide variety of numerical methods have been developed for dealing with quantum problems. There are far too many such methods for us to even mention all of them in this chapter. Our goal instead will be to describe a few of the algorithms that have been developed with quantum mechanics in mind, and use them to treat several representative problems.¹ We also hope to find some enlightening overlap with problems and techniques we have encountered in earlier chapters.

This chapter opens with a brief review of the Schrödinger equation, and a few of the fundamental ingredients of quantum theory. We then consider several methods for treating time-independent problems. While we have dealt with partial differential equations in a number of previous settings, the problems associated with the Schrödinger equation involve extra complications associated with boundary conditions and require some extensions of the methods we have previously employed. We also describe how Monte Carlo methods can be teamed up with the variational principle to calculate wave functions and eigenvalues. In the second half of this chapter we turn to time-dependent problems and describe several numerical approaches. The goal, as usual, is to learn how to deal with problems that are difficult or impossible to treat analytically. However, we will find it useful to apply our numerical approaches to several exactly soluble problems, as this will allow us to test the methods and also illustrate some of the key themes of quantum theory.

10.1 TIME-INDEPENDENT SCHRÖDINGER EQUATION: SOME PRELIMINARIES

The time-independent Schrödinger equation for a particle in three dimensions is

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\vec{r})\psi = E\psi, \quad (10.1)$$

where $\hbar$ is Planck's constant, m is the mass of the particle, V is the potential energy, E is the energy of the particle, and ψ is the wave function. As is well known with quantum theory, it is much easier to write this equation down than to really understand it. The key quantity here is the wave function, which has no direct classical counterpart. In the most general case, ψ is a complex function

¹It is not our intention that this chapter be a substitute for a first course in quantum mechanics, although we will review a few aspects of the theory in the next several pages

of position and, when properly normalized, $\psi(\vec{r})^* \psi(\vec{r})$ is the probability per unit volume that the particle will be found at $\vec{r}$. Here we use the usual notation in which ψ^* is the complex conjugate of ψ .

The fact that ψ can be complex will complicate our numerical treatments.² However, another aspect of (10.1) will turn out to be much more important in numerical attacks on this equation. This is a partial differential equation whose form is not very different from the wave equation encountered in Chapter 6, with one important exception. The energy E is *also* an unknown. A complete solution involves determining both ψ and the corresponding energy. These are also known as the eigenfunction and eigenvalue of the equation.³ An interesting property of eigenvalue problems is that in many cases solutions exist only for certain special values of the eigenvalue, in this case E . Mathematically, this is the origin of the discrete energy levels of quantum theory. It is useful to illustrate these points with an example that can be solved analytically; we will revisit this problem shortly using a numerical approach. Consider a particle moving in free space, that is, a region in which the potential is constant. Since we are free to shift the origin of the potential energy scale, we will take $V = 0$ everywhere. For simplicity we also assume that space is one-dimensional so that $\psi = \psi(x)$ and the ∇^2 operator in (10.1) reduces to the second derivative d^2/dx^2 . The Schrödinger (time-independent) equation then becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi, \quad (10.2)$$

which has the general solution

$$\psi = A \exp(\imath kx), \quad (10.3)$$

where $\imath \equiv \sqrt{-1}$, and A and k are constants. That this is indeed a solution to (10.2) can be verified by substitution, which yields

$$E = \frac{\hbar^2 k^2}{2m}. \quad (10.4)$$

The wave function (10.3) has the form of a plane wave with wave vector $k = 2\pi/\lambda$, where λ is the wavelength of the particle. k is also closely related to the particle's momentum, p , with $p = \hbar k$.

In this particular problem there is a solution to the Schrödinger equation for any value of k , and thus all nonnegative values of E are allowed. We do not have quantized energy levels in this example (at least not yet). The constant A is not determined directly by the Schrödinger equation. Our solution satisfies the wave equation (10.2) for any value of A , a fact we should have expected since the equation is linear. Constants such as A , which set the overall magnitude of the wave

²While the wave function may be complex, there are many cases in which its phase can be chosen so as to make ψ real. In fact, in our numerical work we will not encounter any situations that absolutely demand that ψ be complex until we deal with time-dependent problems.

³Equations in which an operator, such as the ∇^2 operator in (10.1), acting on a function is equal to a constant times the function are known as eigenvalue equations and arise in many other contexts, a few of which are described in Chapter 11. See also Appendix H. Griffiths (1995) and Schiff (1968) contain nice introductions to this topic in the context of quantum mechanics.

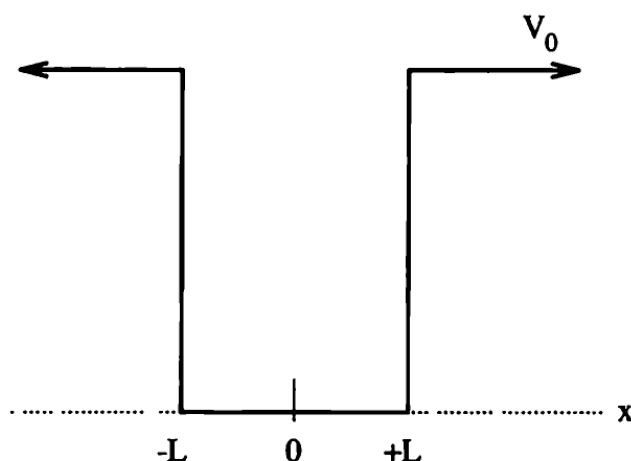


FIGURE 10.1: Potential energy for a particle in a box. The potential is $V = 0$ inside the box (i.e., for $|x| \leq L$) and V_0 outside. If $V_0 = \infty$, then the box is said to have "hard" walls, while if V_0 is finite, the walls are "soft."

function, are determined by the physical interpretation of ψ and its relation to the probability density. We already noted that the probability of finding the particle in a particular region of space is proportional to $\psi^*\psi$. For our one-dimensional case $\psi(x)^*\psi(x)dx$ is the probability of finding the particle within a region of length dx in the neighborhood of x . Since we are considering the behavior of a single particle, this probability must satisfy the normalization condition

$$\int \psi^*\psi dx = 1. \quad (10.5)$$

In words, the total probability for finding the particle somewhere must be unity. This condition can generally be used to set the overall magnitude of the wave function. However, applying it to our plane wave yields a minor difficulty. For the wave function (10.3) we have $\psi^*\psi = A^2$, so that $\int \psi^*\psi dx = A^2 \int dx$. If our particle is allowed to roam freely over an infinitely large region, we must make A infinitely small in order to preserve normalization.

It is convenient to deal with this situation by assuming that the particle is confined to a box that extends from $x = -L$ to $x = +L$. The plane wave (10.3) must then fit in this box. The corresponding potential is now $V = 0$ inside the box and very large outside, as sketched in Figure 10.1. Since $V(x)$ is a discontinuous function of position, it is simplest to solve for ψ separately in the regions $|x| \leq L$ (inside the box) and $|x| > L$ (outside the box). Inside we have $V = 0$, so ψ in this region is still given by (10.3). If we assume that the potential outside is $V = \infty$ and that the energy of the particle is not infinite, then the only way we can satisfy the Schrödinger equation in the outside region is if $\psi_{\text{outside}} = 0$. This implies that $d^2\psi_{\text{outside}}/dx^2 = 0$ also, so both sides of (10.2) are zero. Physically this result should not be surprising. A particle with a finite energy will have no chance of being found in a region where $V = \infty$, so the probability amplitude ψ must vanish.

To complete our solution we make use of one more constraint on ψ ; it must be a continuous function⁴ of x . Since ψ vanishes for $x > \pm L$, the solution for ψ inside the box must vanish at $x = \pm L$. This constraint can be satisfied by noting again that the Schrödinger equation is linear, so the sum of two solutions is also solution. We can therefore construct wave functions of the form

$$\psi_+ = \frac{A}{2} [\exp(ik_+x) + \exp(-ik_+x)] = A \cos(k_+x), \quad (10.6)$$

$$\psi_- = \frac{A}{2} [\exp(ik_-x) - \exp(-ik_-x)] = A \sin(k_-x).$$

These are just standing waves, similar to what is found with a vibrating string. The condition $\psi(\pm L) = 0$ requires that

$$k_+ = \frac{\pi}{2L}, \frac{3\pi}{2L}, \dots = \frac{(2n-1)\pi}{2L} \quad (10.7)$$

$$k_- = \frac{\pi}{L}, \frac{2\pi}{L}, \dots = \frac{n\pi}{L}. \quad (10.8)$$

This constraint on k means that we now have discrete energy levels. Only certain special values of the wave vector are allowed. These values are those yielding standing waves that satisfy the boundary conditions on ψ at the walls of the box, as shown in Figure 10.2. Quantization of k also implies quantization of the energy since $E = \hbar^2 k^2 / 2m$. Constraints such as these boundary conditions generally lead to quantized levels. We will find that they play a major role in constructing numerical solutions to the Schrödinger equation.

It is interesting to notice that the wave functions (10.6) all have a definite parity; they are either even $\psi(+x) = \psi(-x)$, or odd $\psi(+x) = -\psi(-x)$, with respect x . This property can be traced to the potential energy. Whenever the potential is an even function of x , the wave functions can always be written so as to have a definite parity. This will prove useful in developing a numerical approach in the next section. Note also that the wave functions (10.6) can be properly normalized without coefficients that are infinitely small. The normalization condition (10.5) can be used to determine the value of A for these functions, yielding $A = L^{-1/2}$.

Two other features of the wave functions (10.6) will turn out to be very useful later in this chapter. The forms of these functions may remind you of the sines and cosines encountered in connection with Fourier transforms in Chapter 6 and Appendix C. It is thus not surprising that these wave functions form a complete set of orthogonal functions over the interval $|x| \leq L$. The term "orthogonal" means that $\int \psi_1^* \psi_2 dx = 0$ unless ψ_1 and ψ_2 refer to the same eigenstate. The term "complete set" means that any function⁵ in this interval that vanishes at $x = \pm L$ can be written as a sum of these wave functions with suitably chosen coefficients. The orthogonality property makes this expansion, that is, the set of coefficients, unique. Such a collection of functions is usually referred to as a *basis set*. The similarity to Fourier transforms, in which the basis set is also composed of sines

⁴This condition is required to make quantities such as the momentum physically well behaved.

⁵We will not worry about pathological functions here

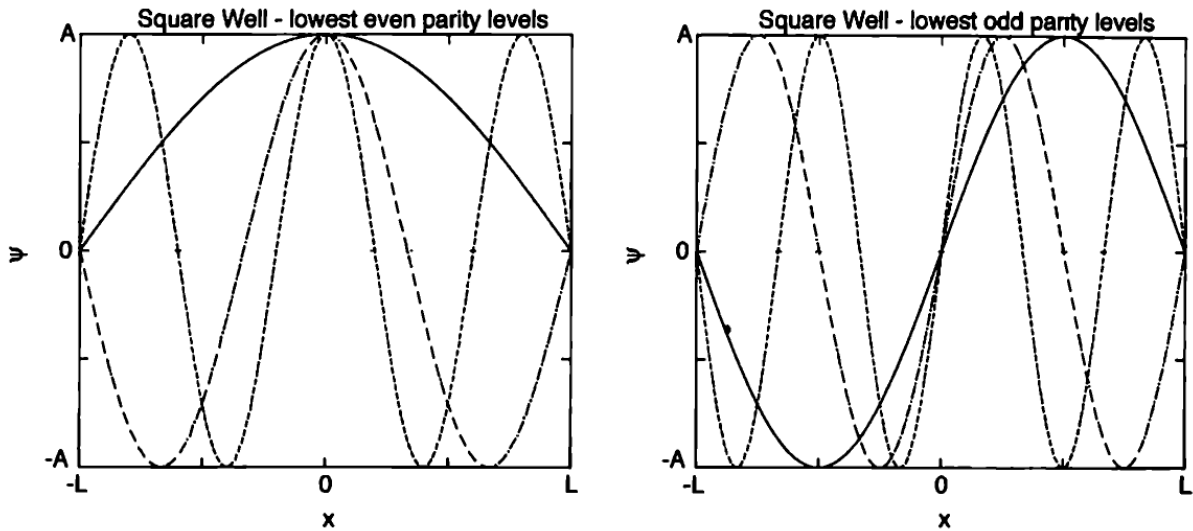


FIGURE 10.2: Wave functions for the lowest six eigenstates for a particle in a box. These are the standing waves in (10.6). The figure on the left shows the cosine-like (even parity) standing waves, while the sine-like (odd parity) wave functions are shown at right. In each case the solid curve is the lowest energy state (of that parity). Note that $A = L^{-1/2}$ is required for the wave functions to be properly normalized.

and cosines, should be evident. However, a basis set need not be this simple, as we will see in an example later in this chapter.

10.2 ONE DIMENSION: SHOOTING AND MATCHING METHODS

We next consider how we can solve numerically the time-independent Schrödinger equation in one dimension, using methods similar to those we have previously developed in connection with ordinary differential equations. We begin with the particle-in-a-box problem, as this will allow us to compare with the analytic solution obtained above. We will then consider several problems for which analytic solutions are not available.

The Schrödinger equation in one dimension (10.2) is very similar in form to the equations of motion we encountered in connection with projectiles in Chapter 2. In those cases we had a differential equation involving the position x , as a function of time. Starting with initial values for x and dx/dt we were able to numerically integrate forward in time to obtain $x(t)$. Now we have a differential equation for ψ as a function of x and we want to use a similar approach. To do this we need the values of ψ and $d\psi/dx$ at some location so that we can begin the integration. These “initial conditions” for the wave function and its derivative are analogous to the initial conditions in a projectile problem. For a projectile these conditions are determined by how the object is struck or thrown, etc. The initial conditions for a wave function are obtained in a somewhat different manner.

When the potential is symmetric, as it is for our particle in a box, we can use the symmetry to our advantage. We noted earlier that if $V(x)$ is an even function of x , the wave functions can be written as purely even or purely odd functions of x . That is, we can choose the wave function to satisfy either⁶ $\psi(+x) = \psi(-x)$, or $\psi(+x) = -\psi(-x)$. Indeed, we have already observed this symmetry in the analytic solutions for the infinite square well, Figure 10.2. An even parity solution has $d\psi/dx = 0$ at $x = 0$, and it is safe to assume (as we will see shortly) that $\psi(0) \neq 0$. Since the Schrödinger equation is linear, we also know that a solution can be multiplied by a constant factor and still be a solution. We are therefore free to *provisionally* choose $\psi(0) = 1$ as an initial value for integration of the Schrödinger equation. When we are finished and have a numerical solution of the equation in hand, we can then normalize that result by a constant factor so as to satisfy the probability constraint $\int \psi^* \psi dx = 1$ and thereby obtain a physically acceptable wave function.

The same approach can be used to obtain an odd-parity solution. Since in this case $\psi(x) = -\psi(-x)$, we must have $\psi(0) = 0$ and $d\psi/dx \neq 0$ at $x = 0$. We are again free to pick the scale of ψ , as long as we normalize it at the completion of the calculation, so we can use $\psi(0) = 0$ and $d\psi/dx = 1$ as initial conditions.

By exploiting the symmetry we have thus obtained values for ψ and $d\psi/dx$ at $x = 0$. These are the initial conditions for our integration. The next step is to rewrite (10.2) in finite-difference form. In most of our previous examples we have written such second-order differential equations as two separate first-order equations. Here, since we aren't interested in the derivative $d\psi/dx$, we will take a slightly different approach (which may remind you of the Verlet method). As usual, we discretize space in steps of size Δx and write $\psi_n \equiv \psi(n\Delta x)$. The second derivative in (10.2) can be written in the usual finite-difference form

$$\frac{d^2\psi}{dx^2} \approx \frac{\psi_{n+1} + \psi_{n-1} - 2\psi_n}{(\Delta x)^2}, \quad (10.9)$$

which is similar to the expression we encountered in connection with the wave equation in Chapter 6. Inserting this into (10.2) we have

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} \approx -\frac{\hbar^2}{2m} \left[\frac{\psi_{n+1} + \psi_{n-1} - 2\psi_n}{(\Delta x)^2} \right] \approx (E - V_n) \psi_n, \quad (10.10)$$

where we have written $V_n = V(n\Delta x)$ and have used the $\approx$ signs to emphasize that this is only an approximation to the original differential equation. This can be rearranged to obtain ψ_{n+1} in terms of ψ_n and ψ_{n-1} . For convenience we will now follow the usual convention for such problems and employ units in which $\hbar = 1$ and $m = 1$, which leads to

$$\psi_{n+1} = 2\psi_n - \psi_{n-1} - 2(\Delta x)^2 (E - V_n) \psi_n. \quad (10.11)$$

Let us first consider the even-parity solutions. We have already argued that we can take $\psi(0) = 1$, so we will first apply (10.11) at $n = 0$ (i.e., $x = 0$). Since

⁶If, in addition, the eigenvalue E is non-degenerate (i.e., if there is only one linearly independent solution with this value of E), then the eigenfunction $\psi(x)$ for E must possess a definite parity.

we also have $d\psi/dx = 0$ at $x = 0$, we take $\psi_{-1} = 1$ as well, so we can use (10.11) to calculate $\psi_{n+1} = \psi_1$. We then move on and use ψ_0 and ψ_1 to obtain ψ_2 . The process can be repeated to find ψ_n for all values of n that are of interest.

This procedure is similar to the way we dealt with projectile motion. However, we now have a complication that we didn't have with baseballs. From the analytic solution for a particle in a box we saw that with hard walls ($V_0 = +\infty$) the wave function must vanish at the walls. This means that we have a boundary condition that ψ must satisfy at $x = \pm L$. In terms of projectiles this is analogous to requiring that the baseball land at a precise location. This will only happen if the initial velocity of the ball has a particular value, or perhaps a value chosen from a discrete set of possibilities. Likewise, we will see that for our wave function the conditions at $x = \pm L$ will be satisfied only for certain values of the energy.

It is useful at this point to consider, in a qualitative sense, the behavior of ψ_n obtained by iterating (10.11) with different values of E . Schematic results for three different values of the energy near the ground-state value E_G (you should recall that the ground state is an even-parity solution) are shown in Figure 10.3. We already know E_G from the analytic results. For the units we have chosen here ($\hbar = 1$, $m = 1$, and a box with $L = 1$), the ground state has an energy⁷ of $E_G = \pi^2/8 = 1.2337\dots$. If we were to use (10.11) to calculate ψ for a value of E that is smaller than this, we would find that ψ is greater than zero when we reach $x = L$ and then diverges to $+\infty$ at larger x , so the boundary condition is *not* satisfied. This is illustrated by the top curve in Figure 10.3. While this function is a solution of (10.2), it cannot be normalized and thus cannot be made to yield a physically acceptable probability density. It is therefore not an allowed wave function for our particle. On the other hand, if we were to perform this calculation with a value of E that is slightly larger than E_G , we would find that ψ drops too quickly with x , so that it is negative when we reach $x = L$ and diverges to $-\infty$ at larger x , as illustrated in Figure 10.3. Only if E has *precisely* the value E_G will the wave function reach zero at precisely $x = L$ and remain zero at larger values of x . This wave function is normalizable and is the ground state ψ for our particle in a box.

This approach for solving a differential equation that has boundary conditions that must be satisfied at *both* ends of an interval is known as the *shooting method*. The method is analogous to trying to throw a baseball or shoot a cannon shell so as to hit a specified target. Only for certain special conditions, in this case the value of E , will the function satisfy the required boundary conditions (that is, will the ball land at the desired spot). In the present case the boundary conditions⁸ are such that the wave function must vanish at two particular locations ($x = \pm L$). This was a result of choosing the potential outside the box to be infinitely large. We will see in a moment how to handle cases in which the potential energy varies more slowly with position, such as a box with "soft" walls or a harmonic oscillator.

Let us next consider how to construct a program that implements this approach. The basic plan is to use (10.11) to calculate $\psi(x)$ for different values of the energy, and choose the result that comes closest to satisfying the boundary

⁷This can be verified from the results for k_+ (10.7) in the previous section

⁸Because of the symmetry, we in effect have boundary conditions at $x = 0$ and $x = L$.

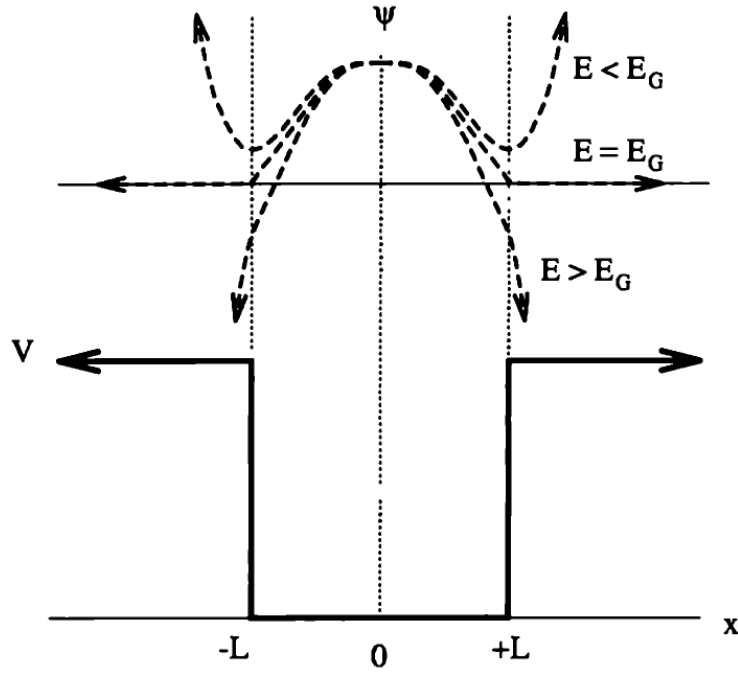


FIGURE 10.3: Schematic results for ψ calculated from (10.11) for several values of E . Top: behavior for the even-parity solution with E near the ground-state energy, E_G . Bottom: corresponding potential energy. The potential outside the box ($|x| > L$) is assumed to be very large.

conditions at the walls of the box. This procedure will yield *both* the wave function and the corresponding energy. We will actually design our program to search systematically for the proper value of E and thereby approach the correct value very closely. The routine sketched below uses the shooting method to integrate starting from $x = 0$ (we again assume $L = 1$), beginning with $\psi_{-1} = \psi_0 = 1$ corresponding to an even-parity solution.⁹ After initializing variables such as the spatial step size Δx and the initial estimate for the energy E , the subroutine `calculate` is called. This routine integrates from $i=0$ (which corresponds to $x = 0$) to larger values of i , stopping when the wave function ψ begins to diverge. At this point the loop is exited and the results for ψ can be displayed. From Figure 10.3 we know that if E is not precisely equal to one of the eigenvalues, the value of the energy should now be adjusted either up or down depending on which direction ψ last diverged. Our routine adds an amount ΔE to E , giving a new estimate for the energy, and the integration is repeated. Let us suppose that E has just been increased ($\Delta E > 0$). If in the next trial ψ diverges in the *same* direction as in the previous case, this means that we did not change the energy enough.¹⁰ The

⁹We will leave it to the exercises to modify this routine to deal with the odd-parity solutions.

¹⁰This assumes that the last variation in E was made in the *correct* direction. Once the procedure starts to home in on a particular eigenvalue, having reversed the sign of ΔE once or more, this assumption is always true. However, initially, before the sign of ΔE is reversed the first time, it may not be true and the procedure may run away looking for an eigenvalue lower than the ground state energy E , or else home in on an eigenvalue different from the one we are after.

routine would then increase E again and repeat the integration. However, if ψ now diverges in the opposite direction, that means we have changed E by too much and have *overshot* the correct value of the energy. The `calculate` routine then cuts the value of ΔE in half, changes its *sign*, and adds it to E , so that we now proceed back toward the solution. We thus hunt down the correct value of the energy with ever-decreasing values of $|\Delta E|$.

EXAMPLE 10.1 **Interactive shooting method routine `calculate`**

- Input are the number N and the size Δx of the spatial grids, the initial guess for the energy E and increment ΔE , and the cutoff parameter b (see below).
 - Implement $\psi'(0) = 0$ for an even parity solution by setting $\psi_0 = \psi_{-1} = 0$.
 - Initialize variable *last_diverge* (which keeps track of the direction of the diverging trend) to zero (since we do not know this direction *a priori*). If the subsequent calculations to evaluate ψ for a given value of E exits by detecting a diverging trend, the sign of the last value of ψ evaluated indicates the direction.
 - Main loop in which we iterate and home-in to a valid eigenvalue E :
 - ▷ For the current value of E , loop through the index $i = 0, 1, \dots, N - 1$ to successively compute the wave function $\{\psi_i\}$.
 - Calculate ψ_{i+1} from ψ_i and ψ_{i-1} using

$$\psi_{i+1} = 2\psi_i - \psi_{i-1} - 2(E - V(x_i))(\Delta x)^2\psi_i.$$
 - Check if $|\psi_{i+1}| > b$ for a suitable cutoff value b .
 - ▷ If yes, assume that ψ is diverging. Exit the loop over i and display the computed ψ .
 - ▷ Otherwise, continue the loop with the next value i .
 - ▷ Display the computed ψ and current value of ΔE , and wait for key input to see if we should continue further.
 - If ΔE is small enough, we may conclude that the current E is acceptable (if the displayed ψ looks good), or reject this calculation (if it is not). Exit the subroutine.
 - If further iteration is indicated, continue.
 - ▷ If the last value of ψ evaluated ($\psi(i+1)$) and *last_diverge* are of different signs, turn around the direction of varying E and halve its magnitude by assigning $\Delta E = -\Delta E/2$.
 - ▷ Update the trial value of E to $E + \Delta E$.
 - ▷ Replace *last_diverge* by the sign of $\psi(i + 1)$.
 - ▷ Go back to the beginning of the homing-in process and iterate.
-

In the `calculate` routine it is convenient to employ a user-defined function¹¹ for $V(x)$. Using a custom function in this way makes `calculate` easier to read and understand and also makes it simpler to modify the program to deal with other potential functions without modifying any other parts of the program. The *cutoff* parameter b is used to judge whether $\psi(x)$ is diverging (with no hope of return); the appropriate value of b depends on the eigenvalue E as well as the type of potential $V(x)$. For the square-well problem treated here and the choice of $\psi(0)$ (or $\psi'(0)$ for the odd solutions), a value of $b \approx 2$ works well, but you need to experiment with it for other situations.

Also to be noted is the interactive nature of this routine. When $\psi(x)$ for some trial value of E has been computed (or when $|\psi_{i+1}| > b$ for some i), it is useful to display the current $\psi(x)$ and asks the user whether to continue. This is an optimization problem (similar to the one we discussed in connection with estimating the launch angle that maximizes a cannon range in Chapter 2, for example), and a visual evaluation of how good (or bad) the current wave function is can be valuable, particularly at initial stages of exploring possible solutions. Most programming languages allow some kind of interactive polling from the keyboard. On the other hand, it is not hard to convert this program into an automatic mode, in which case one needs to set a stopping criterion; setting a limit on the magnitude of ΔE is usually a good approach.

Some results from this program are shown in Figure 10.4 where we plot ψ calculated for several values of E near the exact value for the lowest energy level, $E_G = 1.2337\dots$. Here we have taken $V = 0$ inside the box and $V = V_0 = 1000$ outside. This behavior is just like the schematic illustrations in Figure 10.3, with ψ diverging to either $+\infty$ or $-\infty$, depending on whether E is above or below the energy that best satisfies the boundary conditions. As the search routine approached this value, the boundary condition $\psi(L) = 0$ was satisfied with increasing precision and the divergence of ψ was pushed to larger and larger values of x . The last value of E considered here was 1.1790; decreasing it by only 0.0001 caused ψ to diverge to $-\infty$. Hence, these two values of E closely bracket the “best” numerical value. You may notice that this is a few percent lower than the exact value for a box with infinitely hard walls. We will see in a moment that this difference is due (largely) to the value used for V outside the box. The exact value for E_G quoted above assumes $V_0 = +\infty$, while the calculation here used $V_0 = 1000$.

This procedure can be employed to find the other even-parity levels by beginning the search at higher values of E . Finally, we should point out that the wave functions given in Figure 10.4 are not properly normalized. To do this would require that ψ be rescaled by a constant factor so as to satisfy (10.5). Note that this rescaling should be done using the wave function for $|x| \leq L$ from Figure 10.4 (that is, the diverging part outside the box should not be included). We will leave this task to the reader.

Next we consider the calculation of the odd-parity levels. We will leave this job mainly to the exercises, although we will show a few results here. As we

¹¹By this we mean a function that is constructed by the user as opposed to being built into the language (or its libraries)

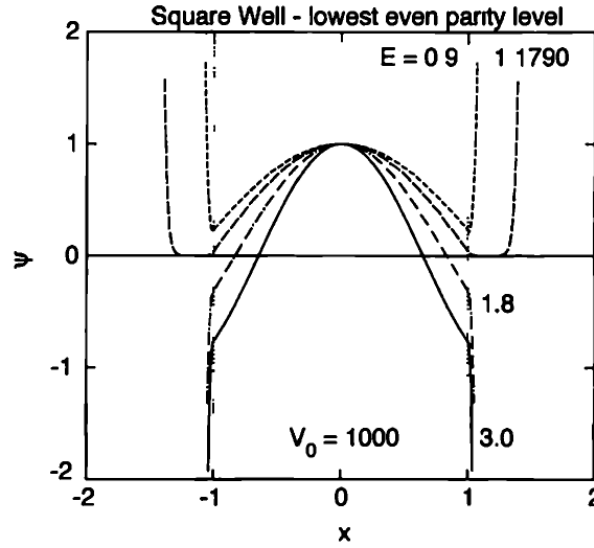


FIGURE 10.4: Trial ground-state wave functions calculated for a box, that is, a square well, with the shooting method for several values of the energy. The potential was $V = 0$ inside the box ($|x| \leq 1$) and $V = 1000$ outside. The dotted lines show the positions of the walls of the box. The spatial step size was $\Delta x = 0.01$. The calculation assumed $\psi(0) = 1$, so the wave functions are not normalized.

have already mentioned, the odd-parity wave functions are the solutions to (10.2) for which $\psi(+x) = -\psi(-x)$. Applying this condition at $x = 0$ requires that $\psi(0) = 0$. Since, as we have argued above, the overall magnitude of ψ at this stage is arbitrary,¹² we take $d\psi/dx = 1$ at $x = 0$. Hence, the calculation routine can again be used to obtain $\psi(x)$ by integration, the only difference being that we must employ the initial values $\psi_{-1} = -\Delta x$ and $\psi_0 = 0$. The rest of the calculation proceeds as before, as our routine hunts down the value of E that best satisfies the boundary conditions at $x = \pm L$. Some results for several different values of E are shown in Figure 10.5; the value we find for the energy of the lowest odd-parity level is ≈ 4.670 , which is about 5 percent below the exact value for a box with infinitely high walls ($\pi^2/2 = 4.934\dots$). As with the ground-state energy, this difference is due mainly to the fact that the potential outside the box is not infinitely large.

If you were to examine Figures 10.4 and 10.5 with very high resolution, you would notice that while the wave function at the walls of the box is small, it is not quite zero, even for the best values of E . You may recall that we obtained the boundary condition $\psi = 0$ at the walls of the box under the assumption that the potential is infinite outside the box. Actually, in our computations we have used a slightly different criteria for choosing the best wave function. This criteria involved the divergence of the wave function at large x and is a more general approach since it can be applied to a wide variety of potentials (as we will see in a moment). For a box with infinitely hard walls, that is, if the potential outside the box is truly infinite,

¹²We can once again adjust the magnitude of $\psi(x)$ at the end of the calculation so as to satisfy the normalization condition $\int \psi^* \psi dx = 1$.

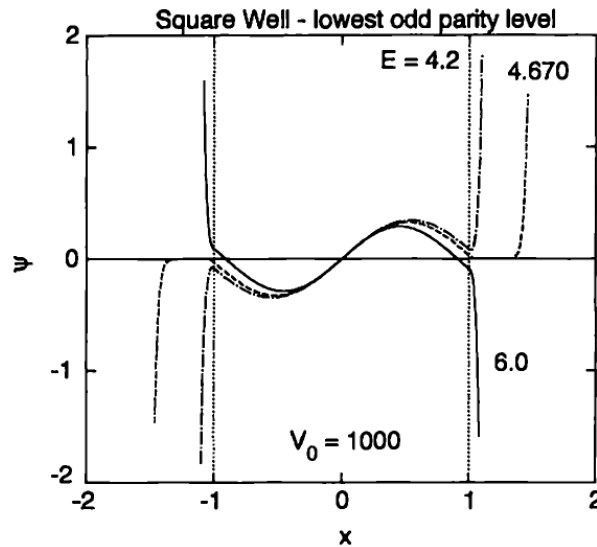


FIGURE 10.5: Trial wave functions for a square well with $V_0 = 1000$. All of the parameters are the same as in Figure 10.4, except that here we consider the lowest odd-parity level.

this divergence criteria is equivalent to the boundary condition that ψ vanish at the edges of the box. However, for a box with soft walls the wave function will be small, but nonzero, in the region beyond the walls. This is a well-known result of quantum theory, which is not found in classical physics.

According to classical theory, a particle cannot venture into a region where V is larger than its total energy. This would require that the classical kinetic energy be negative, which is not allowed. However, quantum theory places no such restriction on the particle. There is a small, but nonzero, chance of finding the particle *anywhere* in space so long as the potential energy is not infinite. This leads to a variety of interesting and important effects such as tunneling, which we will explore shortly. For the particle in a box problem it means that ψ need not go completely to zero at the walls of the box. It turns out that when $V > E$ the wave function decays exponentially with distance as we go farther and farther outside the box. This decay becomes more rapid as $V_{\text{outside}} \equiv V_0$ is made larger, as illustrated in Figure 10.6. Here we show the wave function for the lowest energy level with different values of V_0 . In all three cases E is within ± 0.0001 of the ground-state energy for that particular value of V_0 . We see that the magnitude of ψ at the wall increases as V_0 is reduced. In addition, the rate at which ψ diverges as $|x|$ is increased beyond the walls (at $x = \pm 1$) also increases greatly as V_0 is made larger. We will explore this behavior further in the exercises and in some other examples below.

The shooting method is a convenient way to deal with boundary-value problems in one dimension. However, it is limited to situations in which values of ψ and $d\psi/dx$ at some initial location can somehow be determined. In our particle-in-a-box problem the symmetry of the potential was essential for obtaining ψ and its deriva-

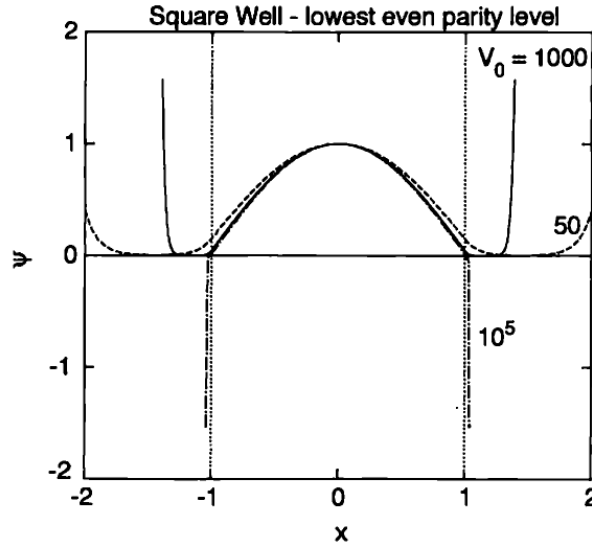


FIGURE 10.6: Ground-state wave function (unnormalized) calculated using the shooting method for a square well with different values of the potential outside the box, V_0 . The walls of the box are at $x = \pm 1$. The ground-state energies here are 1.019, 1.179, and 1.221 for $V_0 = 50, 10^3$, and 10^5 . The spatial step size was $\Delta x = 0.01$.

tive at the center of the box. In situations involving a nonsymmetric potential, the wave functions will not have a definite parity; thus we cannot obtain initial values of ψ and $d\psi/dx$ from symmetry arguments. In such cases a different approach that relies on the continuity of ψ can be employed. Consider the nonsymmetric potential shown schematically in Figure 10.7. This has a form much like the Lennard-Jones potential we encountered in our molecular dynamics work in Chapter 9 and is similar to what we might find for a system involving two atoms, with one located¹³ at $x = 0$. Note, however, that in this example we restrict things to one dimension, so we are not yet treating the full three-dimensional Schrödinger equation. We expect on physical grounds to find a bound state in which the atom has a large probability of being found near the minimum of the potential. The wave function for this state will vanish at both large and small x and be significantly different from zero only in the region where V is large in magnitude and negative.

Let us therefore consider calculating ψ using (10.11), with integrations beginning in the regions where ψ is very small (i.e., at large x and small x) and proceeding toward the middle. That is, we now perform two integrations. One starts from a point on the far left in Figure 10.7, at the point $x = x_L$. At this location we set $\psi = 0$ and give the derivative $d\psi/dx$ a very small value. We then use (10.11) to obtain ψ at $x_L + \Delta x$, $x_L + 2\Delta x$, etc., and thus determine ψ at ever-increasing values of x . We call this function ψ_L , since it is obtained by integrating in from the

¹³For simplicity we can imagine that the atom at $x = 0$ is much more massive than the other so that it can be taken as fixed in space, and we need worry only about the wave function of the light atom.

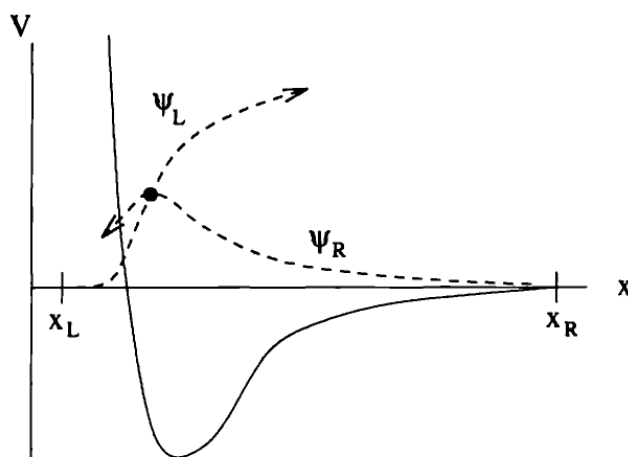


FIGURE 10.7: Schematic illustration of the matching method for solving the Schrödinger equation. The solid curve shows the potential, while the dashed curves show (hypothetical) wave functions calculated using (10.11), starting the integration from the left (ψ_L) and from the right (ψ_R). In this case the two trial functions do not match smoothly, and thus cannot be combined to give an acceptable solution of the Schrödinger equation.

left-most region. This procedure is then repeated starting from a point $x = x_R$, far to the right in Figure 10.7. At this location we set $\psi = 0$ and give $d\psi/dx$ a small value, in preparation for the application of (10.11). We then integrate toward smaller values of x , and obtain a function we call ψ_R . The plan is to combine the functions ψ_L and ψ_R to obtain the wave function over the entire range.

The correct wave function must be a smooth, continuous function. That is, ψ and its derivative must both be continuous.¹⁴ This means that if a combination of our two functions ψ_L and ψ_R is to form an acceptable wave function, they must intersect somewhere (so that the wave function is continuous), and they must have the same slope at the intersection point (so that $d\psi/dx$ is continuous). We can always arrange for ψ_L and ψ_R to intersect since, as has already been noted several times, we can scale the calculated ψ by a constant factor and still obtain a solution to the Schrödinger equation. It is usually convenient and numerically advisable, to pick a location somewhere near the minimum of the potential as the matching point since the wave function will be large there, and then scale either ψ_L or ψ_R or both so that their values are equal at that point.¹⁵ This still leaves us with the requirement that the derivatives be equal at the matching point. The only remaining adjustable parameter in the computation is the energy. For most values of E we find the situation depicted in Figure 10.7, where the derivatives $d\psi_L/dx$ and $d\psi_R/dx$ do *not* match. However, it is usually possible to adjust the value of E so as to make these derivatives match. When this occurs, the combination of ψ_L and ψ_R gives the wave function, and the associated value of E is the energy of the level.

This approach is often called the *matching method*, for obvious reasons. It

¹⁴This assumes that the potential is a smooth function.

¹⁵This is equivalent to scaling the derivatives at x_L and x_R so as to make $\psi_L = \psi_R$ at the matching point.

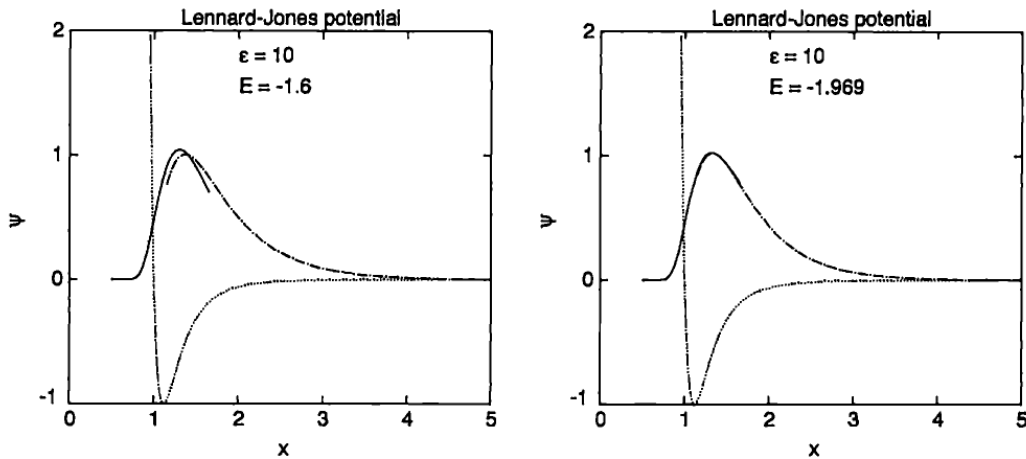


FIGURE 10.8: Solution for the ground state of the Lennard-Jones potential [(10 12) with $\epsilon = 10$ and $\sigma = 1$] constructed using the matching method. The integrations were started at $x_L = 0.5$ (on the left) and $x_R = 5$ (on the right) with $\Delta x = 0.01$. The two trial functions are shown by the solid (ψ_L) and dot-dashed (ψ_R) curves, and the potential is shown by the dotted curve (the scales for ψ and $V(x)$ are not the same). Left: for $E = -1.6$ the derivatives of ψ_L and ψ_R are not equal at the matching point ($x = 1.4$). Right: for $E = -1.969$ the derivatives match fairly well, so this is an acceptable approximation for the wave function. Note that this wave function is not normalized.

can be implemented numerically by modifying the `calculate` routine we developed for the shooting algorithm. A search routine can again be used to obtain an ever more accurate match of the derivatives $d\psi_L/dx$ and $d\psi_R/dx$, and thus better and better estimates for the wave function and energy. While we can use essentially the same structure as the `calculate` routine of the shooting method, the details differ considerably. First, since we are starting to compute $\psi(x)$ from both ends of the system, we must have initial values of the wave functions and their derivatives at both ends. That is, we need to start with $\psi_L(x_L - \Delta x)$, $\psi_L(x_L)$, $\psi_L(x_R + \Delta x)$, and $\psi_L(x_R)$, where the two values outside the interval (x_L, x_R) are needed to start the iteration of (10.11). These values are usually chosen so that the initial slopes are very small, e.g., $\psi_L(x_L - \Delta x) = -0.0001\Delta x$. For convenience we can choose the same value at both ends as they will be scaled differently to match the values of ψ_L and ψ_R at a central point anyway. Second and more importantly, we must have an efficient strategy to vary E so that the derivatives $d\psi_L/dx$ and $d\psi_R/dx$ approach each other. A valid approach here is to: (1) assign a value of ± 1 to a variable, say, *match*, depending on whether $d\psi_L/dx > d\psi_R/dx$ or vice-versa, and (2) if the sign of *match* is different from the corresponding quantity from the previous trial value of E , then (3) assign the new value $\Delta E = -\Delta E/2$ to the increment to use next. We leave it to the reader to confirm that this method works regardless of the signs of the derivatives and to construct a working program based on this strategy (of course, you are also welcome to design your own strategy!).

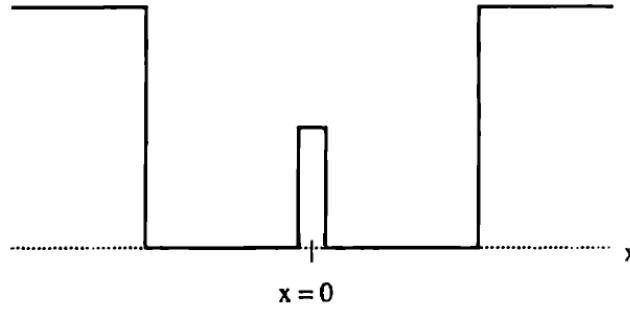


FIGURE 10.9: Two potential wells separated by a narrow "tunnel" barrier

Figure 10.8 shows results for the one-dimensional Lennard-Jones problem¹⁶

$$V(x) = 4\epsilon \left[\left(\frac{\sigma}{x} \right)^{12} - \left(\frac{\sigma}{x} \right)^6 \right] \quad (10.12)$$

Here we plot results for two values of E . In each case we have normalized both ψ_L and ψ_R to unity at the matching point $x_{\text{match}} = 1.4$. For the case on the left there was a sizable discontinuity in $d\psi/dx$ at the matching point, so this solution is not satisfactory. After letting the search routine iterate several times, we obtain the result on the right, where ψ_L and ψ_R join much more smoothly. By starting the search at other values of E we can obtain the wave functions for other energy levels. We will, as usual, leave this for the exercises.

While we introduced the matching strategy for dealing with nonsymmetric potentials, it is not limited to such cases. An interesting example is a square well containing a small potential barrier inside the well, shown schematically in Figure 10.9. This particular potential is actually symmetric, so we could use the shooting method here, starting from $x = 0$. However, there are many interesting variations on this problem; for example, we could move the small barrier off center or make the potential of the barrier vary with x . Such asymmetric potentials require the matching approach.

Before we construct the wave functions for the potential in Figure 10.9, it is worthwhile to anticipate some of the physics of the problem. This potential consists of two square wells, one to the left and one to the right of the barrier. If the barrier is large (very high or very wide or both), the two wells are effectively uncoupled, since the wave function in one will not penetrate far enough into the barrier to reach into the other well (compare with Figure 10.6). The wave functions in each well will therefore be unaffected by the presence of the other, and the solutions will be those we have already encountered in our work on the particle-in-a-box problem. The ground-state wave function for this case is shown schematically on the left side of Figure 10.10.

If, on the other hand, the barrier separating the two wells is not large, the unperturbed wave function from one side will be able to penetrate through to the

¹⁶You may recall that this is the potential we used in our molecular-dynamics calculations in Chapter 9

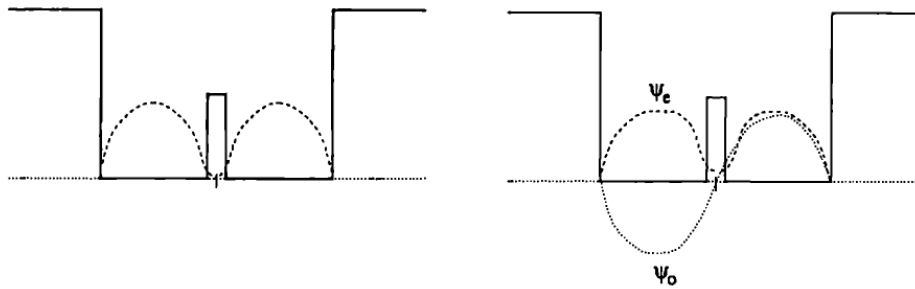


FIGURE 10.10: Left: the dashed curves showing the approximate unperturbed wave functions in the two nearby potential wells, right: wave functions (very schematic!) after we allow for mixing of the unperturbed wave functions

other before it decays to a negligibly small value. Actually, the unperturbed wave functions from both wells will decay as they penetrate the barrier from opposite sides. These two wave functions must meet continuously in the center, just as we saw with ψ_L and ψ_R in the Lennard-Jones problem (Figure 10.8). This leads to what is usually referred to as *mixing* of the two, initial, unperturbed wave functions. If we denote the unperturbed wave functions as ψ_1 and ψ_2 , then to a first approximation they will combine to form wave functions for the entire two-well system as either $\psi_e \sim \psi_1 + \psi_2$, or $\psi_o \sim \psi_1 - \psi_2$. That is, they can form either an even (ψ_e) or odd (ψ_o) combination.¹⁷ These are sketched on the right side of Figure 10.10. Note that the wave functions we have sketched are derived (approximately) from the initial unperturbed wave functions in the two separate wells. Similar combinations are formed from the unperturbed excited-state wave functions.

The two wave functions ψ_e and ψ_o will have energies close to that of the initial unperturbed states. According to perturbation theory, the mixing of two such states leads to a small splitting of the energy levels, with the even level (ψ_e) being pushed to a lower energy and the odd level pushed higher. This can be illustrated nicely using the solutions obtained from the matching algorithm. The ground state has even parity and is considered in Figure 10.11. On the left we show the behavior for a value of E that is too low, and we see that ψ_L and ψ_R do not match smoothly. In the middle case E is slightly too large, although the derivatives of ψ_L and ψ_R match fairly well on the scale shown here. On the right we show results for a value of E that yields a slightly better match. It is interesting that even here the wave function is not quite symmetric; you can see this from the fact that the amplitudes of the two maxima are not quite the same. This is a result of the numerical errors that arise when integrating through a classically forbidden region (the tunnel barrier), as the solutions tend to be divergent in such regions (compare again with Figure 10.6). In this particular problem we could have obtained a better result by employing the shooting method, as this would preserve the symmetry inherent in the problem. However, using the matching method in this case gives a good impression of the strengths and weaknesses of that algorithm.

¹⁷The precise parity of the solutions in this case can be traced to the symmetry of the potential. When the potential is not symmetric the solutions will not have a definite parity, but there will still be two wave functions constructed (approximately) as the sum and difference of the two unperturbed wave functions

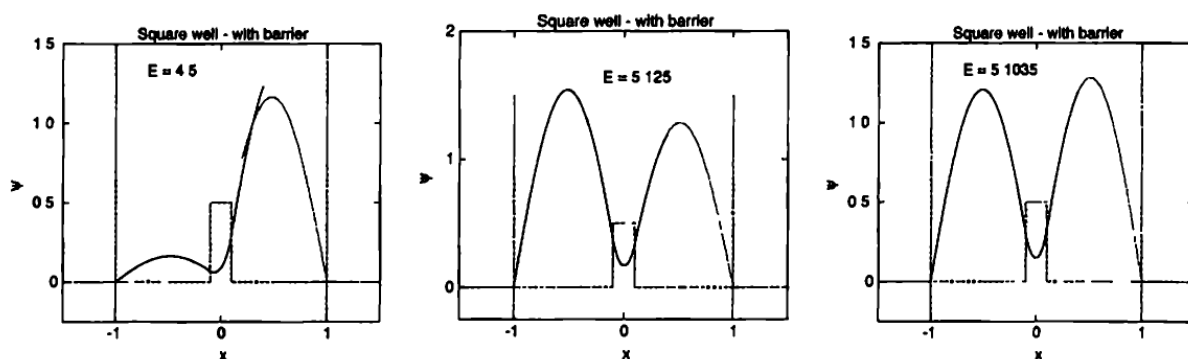


FIGURE 10.11: Results for ψ_L and ψ_R calculated with the matching method for the potential in Figure 10.9, for several values of E . The integrations were started at $x = \pm 1.3$, the step size was $\Delta x = 5 \times 10^{-4}$, and the matching was done at $x_{\text{match}} = 0.3$. The parameters associated with the potential were $V = 0$ inside each well, $V = 10^5$ for $|x| > 1$, and $V = 100$ for $|x| \leq 0.1$. The vertical dashed lines indicate the edges of the potential wells. Note that the scale for ψ varies from plot to plot.

The numerical difficulties associated with integrating through a classically forbidden region (where $V > E$) bring up a related issue. In all of our applications of the matching method we have started the integrations in such forbidden regions and integrated toward the regions where $V < E$, that is, the regions where we expect the wave function to be large. In principle we could have started at a location where $V < E$, and integrated toward the classically forbidden regions. However, the numerical problems and divergences are usually more severe in the latter case. As a general rule it is best to integrate away from classically forbidden regions, toward regions where ψ is large.

Figure 10.12 shows results for slightly higher values of E . The left and middle plots show behavior for values of E that are too low and too high and thus provide poor matches. The plot on the right shows a good match, and the wave function is seen to have odd parity, as expected from the discussion given above.

The shooting and matching methods described in this section are well suited for the time-independent Schrödinger equation in one dimension and similar types of boundary-value problems. We have only been able to consider a few examples here; the exercises explore these and several other cases in more detail.

EXERCISES

- 10.1. Use the shooting method to obtain the lowest six energy levels for a square well. This will require that you write a program that can deal with both even- and odd-parity solutions. Examine how the energy varies with level number and test your values against the exact results (10.7) and (10.8). Compare the wave functions (make sure that they are properly normalized!) with the exact solution (10.6). It is also instructive to examine the behavior as a function of the spatial step size Δx . Study how the value of the ground-state energy varies with the choice of step size; try $\Delta x = 0.05$, 0.01 , and 0.005 , for a box with walls at $x = \pm 1$.
- 10.2. Use the matching method to calculate the wave functions and energies of the

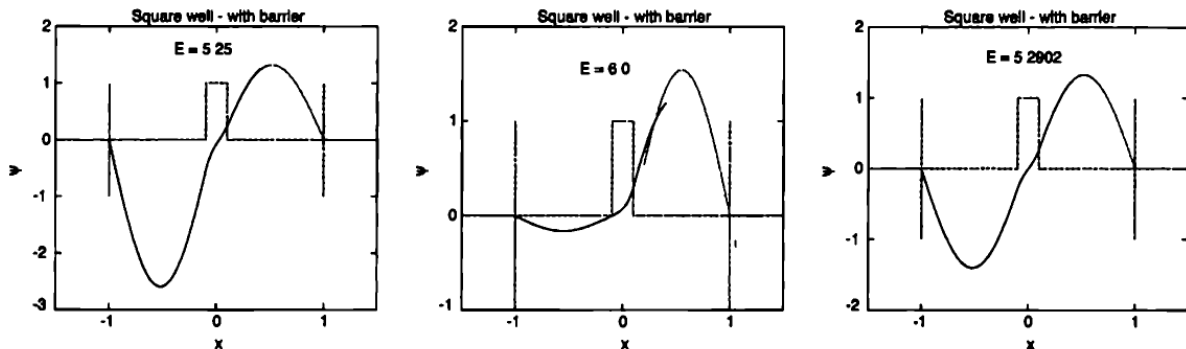


FIGURE 10.12: Results for ψ_L and ψ_R calculated with the matching method for the tunnel potential in Figure 10.9. Here we have used several values of E that are near the energy of the lowest odd-parity level. The other parameters were the same as in Figure 10.11.

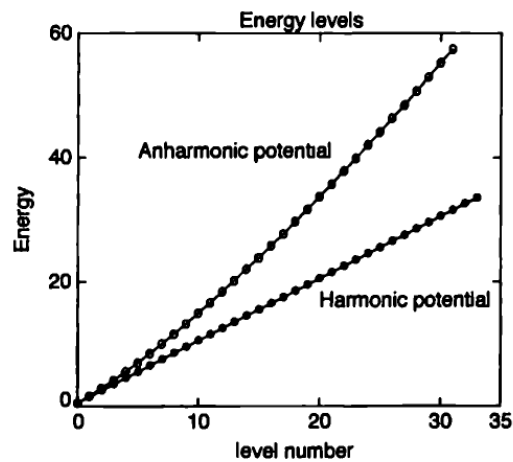


FIGURE 10.13: Lowest few energy levels for a harmonic oscillator ($V = Kx^2/2$ with $K = 1$) and an anharmonic oscillator [$V = (k_1x^2 + k_2x^4)/2$, with $k_1 = 1$ and $k_2 = 0.1$]. In both cases we have used our usual choice of units, for which $\hbar = m = 1$. Note that for the harmonic case the energy varies linearly with level number (which agrees with the well-known, analytic solution). For the anharmonic case the corresponding energies are shifted to progressively higher values. For the higher levels the wave function has more of its "weight" at larger values of x (classically speaking, the vibration amplitude is higher) where the potential is larger due to the anharmonic term, and this pushes the energy higher than in the harmonic case.

first few energy levels of the harmonic oscillator potential $V = Kx^2/2$. Show that the levels are evenly spaced in energy and compare with the exact result $E = (n + \frac{1}{2})\hbar\omega$, where n is an integer (0, 1, 2, ...) and $\omega = \sqrt{K/m}$ (see Figure 10.13). It is also worthwhile to compare the wave functions with the exact results.

- 10.3. Use either the shooting or matching method to calculate the energy levels for the potential $V = x^n$ for different values of n . As n becomes very large this approximates an infinite square well.
- 10.4. Calculate the first few energy levels and the associated wave functions for a potential of the form $V = (k_1x^2 + k_2x^4)/2$. For small x the first (k_1) term dominates, and the behavior is close to that found for the harmonic oscillator, while for large x the second (k_2) term dominates and the behavior is *anharmonic*. Some typical results for the energy levels in this case are given in Figure 10.13.
- 10.5. Use the shooting method to study how the wave function for a particle-in-a-box depends on the magnitude of the potential outside the box V_0 . Examine the variation of ψ beyond the walls of the box and show that it decays exponentially with distance in this region. Study the decay length as a function of V_0 and compare the results for different energy levels. As the energy of the level approaches V_0 the decay length should become larger.
- *10.6. Use the matching method to obtain the wave functions for the first few states above the ground level for the one-dimensional Lennard-Jones potential. Compare the variation of the energy with level number with the results shown for the harmonic and anharmonic potentials in Figure 10.13. Depending on the number of levels you wish to consider, you may have to make the value of ϵ in (10.12) larger than the value used in Figure 10.8. A deeper potential well possesses more bound states.
- *10.7. Many three-dimensional quantum mechanics problems can be reduced effectively to one- or two-dimensional problems. For example, the hydrogen atom is three dimensional as it involves the wave function of an electron as a function of position relative to a (fixed) proton. However, the spherical symmetry makes it possible to write the wave function in the form

$$\psi(r, \theta, \phi) = f(r) Y(\theta, \phi), \quad (10.13)$$

where r is the (radial) distance between the proton and the electron, and θ and ϕ are the usual spherical coordinates [see Griffiths (1995) and Schiff (1968) for discussions of (10.13) and the derivation of (10.14)]. Here $f(r)$ is a function of r only, and all of the angular dependence is contained in the function $Y(\theta, \phi)$. As you may know, these angular functions turn out to be the spherical harmonics. In this problem we will be concerned with the radial function, $f(r)$. When the form (10.13) is inserted into the three-dimensional Schrödinger equation we obtain a differential equation for $f(r)$. It is common practice at this point to deal instead with the related function $g(r) \equiv rf(r)$, which satisfies the equation

$$\frac{d^2g}{dr^2} = \left(2[V(r) - E] + \frac{\ell(\ell+1)}{r^2} \right) g, \quad (10.14)$$

where $V(r) = -1/r$ is the Coulomb potential¹⁸ and ℓ is an integer associated with the angular momentum of the electron that arises from the solution for

¹⁸More precisely, this is the electric potential multiplied by the charge of the electron. We use units in which this charge is -1 .

$Y(\theta, \phi)$. Employ the shooting method to solve (10.14) for $g(r)$ starting your integration at $r = 0$, with the initial value $g(0) = 0$. Consider the cases $\ell = 0$ and $\ell = 1$ and compare your results with the analytic solution for the hydrogen atom. *Hint:* In the units used here the energies of the bound states are given by $-1/2n^2$, where n is an integer.

10.3 A MATRIX APPROACH

The methods developed in Section 10.2 can handle most one-dimensional quantum mechanics problems. However, life is more difficult in two or three dimensions for the following reason. As we have already seen, ψ must, in general, satisfy certain boundary conditions. In one dimension these usually concern the value of ψ at two locations; typically at the two extremes of a region. In higher dimensions we must worry about the value of the wave function on a *surface*, which could be a circle in two dimensions or a sphere in three dimensions. In some cases the potential may have enough symmetry that the problem can be reduced to a differential equation involving only one variable. For example, we have seen in the exercises in Section 10.2 that with the hydrogen atom we can separate the radial and angular dependences of ψ and thereby reduce the radial dependence to a one-dimensional problem. In such cases the matching or shooting method can then be used. Many textbook problems are like this, for the simple reason that more complicated cases can almost never be solved analytically!

Shooting and matching methods are not well suited for satisfying boundary conditions at more than one point and therefore cannot (easily) be used in two or three dimensions. A very general method for dealing with these cases is to write the Schrödinger equation in the form of a matrix eigenvalue problem. While such an approach can be computationally very difficult to carry through, it is useful for illustrating several important points.

Consider a two-dimensional problem and assume that, as usual, we break up space into a lattice of points, with $\psi(m, n)$ being the value of the wave function at location (m, n) on the lattice. The boundary conditions then specify the value of ψ on the boundary. For example, ψ might vanish on the edges of the system; if the system is rectangular, we would have $\psi(0, n) = \psi(M, n) = \psi(m, 0) = \psi(m, N) = 0$, where the lattice runs from $m = 0$ to M and $n = 0$ to N . The Schrödinger equation (10.1) then has the form

$$\begin{aligned} -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r}) \psi &= -\frac{\hbar^2}{2m} \left[\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right] + V(\vec{r}) \psi \\ &\approx -\frac{\hbar^2}{2m} \left[\frac{\psi(m+1, n) + \psi(m-1, n) - 2\psi(m, n)}{(\Delta x)^2} \right. \\ &\quad \left. + \frac{\psi(m, n+1) + \psi(m, n-1) - 2\psi(m, n)}{(\Delta y)^2} \right] + V(m, n) \psi(m, n) \\ &\approx E \psi(m, n), \end{aligned} \quad (10.15)$$

where the last several lines give the wave equation in finite-difference form. This is a system of algebraic equations for $\psi(m, n)$; there are (of order) $M \times N$ unknown values of $\psi(m, n)$ and the same number of equations.

If we define a column vector $\vec{\psi}$ whose elements are $\psi(m, n)$,

$$\vec{\psi} = \begin{pmatrix} \psi(0, 0) \\ \psi(0, 1) \\ \dots \end{pmatrix}, \quad (10.16)$$

and a square matrix $\mathcal{H}$ representing the Hamiltonian operator,

$$\mathcal{H} \equiv -\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}), \quad (10.17)$$

then these equations can be written as a linear equation

$$\mathcal{H} \vec{\psi} = E \vec{\psi}, \quad (10.18)$$

where the Hamiltonian $\mathcal{H}$ is represented by a matrix

$$\mathcal{H} = \begin{pmatrix} 4 + V(0, 0) & -1 & 0 & \dots \\ -1 & 4 + V(0, 1) & -1 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}. \quad (10.19)$$

Here we have taken the liberty to set $\Delta x = \Delta y$ and to absorb factors such as these as well as $\hbar$ and mass m into V and E .¹⁹ The points we wish to make here depend on the general *form* of these equations, so we will avoid the notational distractions associated with specifying all of the factors precisely.

Equations (10.18) and (10.19) define an eigenvalue problem for the Hamiltonian matrix $\mathcal{H}$, which is one of the standard problems in linear algebra.²⁰ Efficient programs for obtaining the eigenvalues E and the associated eigenvectors $\vec{\psi}$ can be found in all serious packages of mathematical routines. Moreover, since $\mathcal{H}$ is usually *sparse*, that is, most of its entries are zeroes, a number of techniques are available to simplify such work.²¹

Focusing on the eigenvalues, these equations are sometimes put in the form

$$\begin{pmatrix} 4 + V(0, 0) - E & -1 & 0 & \dots \\ -1 & 4 + V(0, 1) - E & -1 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} \psi(0, 0) \\ \psi(0, 1) \\ \dots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dots \end{pmatrix}. \quad (10.20)$$

If we define a square matrix $\mathcal{M}(E)$ whose elements are shown as the left-most matrix in (10.20), this can be written as

$$\mathcal{M}(E) \vec{\psi} = \vec{0}, \quad (10.21)$$

¹⁹This procedure is equivalent to setting $\Delta x = \Delta y = 1$, $\hbar = 1$, and $m = 1/2$.

²⁰The reader may note the similarities between this problem and the one encountered in Chapter 7 where we discussed the problem of diffusion on a disordered network as well as the one we will treat in Chapter 11 concerning the vibrations of a membrane. More on the numerical aspects of the linear problems are given in Appendix H.

²¹Nevertheless, computing the eigenvalues and eigenvectors of large matrices is usually a formidable task, which can require large amounts of computer time as we will detail later.

where $\vec{0}$ is a column vector whose elements are all zero. Here we have indicated explicitly the fact that the matrix $\mathcal{M}(E)$ is a function of E . In general there will be MN values of E for which this equation has a nontrivial solution.²² Formally, these eigenvalues are the roots of the algebraic equation

$$\det[\mathcal{M}(E)] = 0, \quad (10.22)$$

called the *secular equation*, where the function $\det[]$ denotes the determinant of a matrix. Each eigenvalue has an associated eigenvector, which is the wave function for that particular energy level.

This matrix approach is completely general and can be very useful, especially in formal arguments. However, it is often a very large computational task unless the Hamiltonian matrix has a special form that reduces the computational requirements. Since the column or row reduction of the determinant would require of the order of $(MN)!$ operations, a naive estimate of the number of required operations might appear to be an exponential function of the matrix size. Fortunately, there are efficient methods available to reduce it to $O([MN]^3)$ (or even lower if the matrix is of a special form such as *tridiagonal*). Sometimes, an approximate solution is sufficient, allowing us to substitute a matrix of a special form (and often a much smaller one as well), saving a large amount of computing resources. However, in general we are still faced with large computational requirements. For example, if there are 50 grid elements along both m and n in a two-dimensional problem, one must deal with a 2500×2500 matrix. In three dimensions the matrix would be $125,000 \times 125,000$. While such eigenvalue problems arise in many applications, and efficient programs for carrying out these computations do exist, computing the eigenvalues and eigenvectors of such large matrices is still a formidable task.²³

We end this section by describing one particularly effective use of the matrix method. This is sometimes called the *power method*, one in which we do *not* explicitly solve for the eigenvalue or the eigenvector, but rely on the repeated application of the $n \times n$ matrix (say, $\mathbf{W}$) on an arbitrary starting trial vector $\vec{u}$ to weed out the contributions to $\vec{u}$ from all but the eigenvector $\vec{u}_1$ with the largest (i.e., "dominant") eigenvalue λ_1 . First, let us assume that the eigenvalues are ordered so that $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$, and assume also that the initial state $\vec{u}$ can be represented as a linear combination²⁴

$$\vec{u} = \sum_{i=1}^n a_i \vec{u}_i, \quad (10.23)$$

where $a_1 \neq 0$ and $\vec{u}_i$ are the eigenvectors with eigenvalues λ_i , so that

$$\mathbf{W} \vec{u}_i = \lambda_i \vec{u}_i. \quad (10.24)$$

²²A trivial solution is one for which all of the $\psi(m, n)$ are zero. Note also that because of the boundary conditions on $\vec{\psi}$, the number of solutions may be less than MN .

²³However, it can be quite useful for quantum problems in which, for some reason, the size of the $\vec{\psi}$ vector (and hence the size of the Hamiltonian matrix) is not too large. Cases of this kind arise, for example, in magnetic problems in which the number of spin (or angular momentum) states is often of order 10 or less.

²⁴This is known to be possible for Hermitian matrices such as those representing a physical Hamiltonian.

Applying $\mathbf{W}$ on $\vec{u}$ then gives

$$\mathbf{W} \vec{u} = \sum_{i=1}^n a_i \lambda_i \vec{u}_i, \quad (10.25)$$

and doing so k consecutive times gives

$$\mathbf{W}^k \vec{u} = \sum_{i=1}^n a_i \lambda_i^k \vec{u}_i = \lambda_1^k \left[a_1 \vec{u}_1 + \sum_{i=2}^n a_i (\lambda_i/\lambda_1)^k \vec{u}_i \right]. \quad (10.26)$$

Since $|\lambda_i/\lambda_1| < 1$ for $i \neq 1$, all but the first term in the square brackets will decrease in magnitude relative to the first term as $k \rightarrow \infty$. Thus, for large k we eventually get

$$\mathbf{W}^k \vec{u} \approx a_1 \lambda_1^k \vec{u}_1, \quad (10.27)$$

and hence $\vec{u}_1$ can be obtained by normalizing $\mathbf{W}^k \vec{u}$.

While this method targets the dominant eigenvalue, there are various extensions which are able to find states other than the one with the largest eigenvalue. For example, if we know that there is an upper bound Λ for the magnitudes of the eigenvalues, we can apply the power method to the modified (and shifted) matrix $\Lambda^2 \mathbf{I} - \mathbf{W}^2$ and obtain the eigenstate with the eigenvalue of the smallest magnitude (say, the ground state wave function). Alternatively, we can also consider the application of the power method with $\mathbf{W}^{-1}$ if we are interested in the ground state. It turns out that this can be accomplished without actually inverting the matrix first. Similarly, it is possible to use the combinations of shifting and inversion to look for eigenstates with eigenvalues in a neighborhood of values of our choice. Moreover, once an eigenstate and its eigenvalue are obtained by such a method, it is possible to iteratively project out the linear space associated with such states and look for the next most dominant one, and so forth. We will not dwell any further on this method, but it can be a very effective way to obtain a specific state, such as the one with the largest (or the smallest) eigenvalue.

10.4 A VARIATIONAL APPROACH

The computational difficulties with a “direct” matrix approach to eigenvalue problems have motivated the development of a number of alternative schemes. We do not have space to discuss all of them here; the interested reader can explore some of them through the references. In this section we will introduce one of these schemes to give an example of how the Schrödinger equation can be attacked. The method we now describe involves a Monte Carlo approach, similar to that described in Chapter 8, coupled with the variational principle.

The variational principle of quantum mechanics can be expressed as follows. Let φ be a “trial” wave function. We imagine that we are searching for the ground-state solution to the Schrödinger equation for a particular problem and that φ is a proposed wave function, or perhaps our best numerical guess for the true wave function. We can calculate the “energy” corresponding to this trial function

$$E^* \equiv \frac{\int \varphi^* \mathcal{H} \varphi d\vec{r}}{\int \varphi^* \varphi d\vec{r}}, \quad (10.28)$$

where the integrations are over all space and $\mathcal{H}$ is the Hamiltonian operator. Note that we have given this effective energy the symbol E^* to emphasize the fact that since φ is not necessarily a solution of (10.17) (it was only a “proposed” wave function), E^* is not necessarily the energy of a true eigenstate.

According to the variational principle, the true ground-state energy is *always* less than or equal to E^* . In fact, the only time they are equal is when the trial wave function is precisely the true ground-state wave function. For *all* other functions φ the effective energy E^* will be higher than the ground-state energy. This principle allows us to recast the problem of solving the Schrödinger equation as a minimization problem. The goal is to find the function φ which minimizes E^* . There are several ways to implement this strategy, corresponding to different ways of searching through the space of all possible wave functions. The approach we describe below is very general, yet also somewhat crude. After seeing how it works, we will consider improved ways of constructing φ and performing the search procedure, some of which will be explored in the exercises.

To be specific we consider again the problem of finding the ground state of the one-dimensional Lennard-Jones potential. We dealt with this problem earlier with the matching method, so we already have a solution with which to compare it. To implement a minimization strategy we must first choose an initial trial wave function, which we will call φ^0 . As usual, we discretize space into bins Δx in size and let φ_n^0 be the value of the initial trial function at $x = n\Delta x$. We also restrict the domain²⁵ of our function to the range $x = 0.7 - 5$ (in reduced units), since we know on physical grounds that the ground state will be one in which the particle is localized near the minimum in the potential, $x_{\min} \sim 1.1$. Hence, our bins run from $n = 0$ corresponding to $x = 0.7$, up to $n = N$ corresponding to $x = 5$. We choose an extremely simple initial trial function: $\varphi_n^0 = \text{constant}$ for $1 \leq x \leq 4$ and zero outside this range, as shown in Figure 10.14. The constant is determined by requiring that our function satisfy the normalization condition $\int \varphi^* \varphi dx = 1$. This is certainly a poor approximation to the true ground-state wave function, so if the method can succeed with this choice of φ^0 it can probably be successful with any choice!

The only remaining part of the algorithm that must be specified is the procedure for how to use φ^0 (or any other trial function) to obtain an improved trial function. We will employ a Monte Carlo approach, which operates as follows. First, we pick one of the bins along x at random by choosing an integer in the range $0-N$; we will call this bin n_{test} . Second, we construct a provisional function by changing the value of the trial function at that location, $\varphi_{n_{\text{test}}}^0$, by an amount chosen randomly in the range $\pm d\varphi$. Third, we calculate E^* for this provisional function. If it is lower than the corresponding value for φ^0 , the provisional function must be “closer” to the true wave function, and we use it as the new trial function. If E^* for the provisional function is higher than the value for φ^0 , we discard the provisional function and keep φ^0 as the trial function.²⁶

²⁵We will take $\sigma = 1$ and $\epsilon = 10$ in (10.12), so that the potential is the same as that considered in Figure 10.8.

²⁶Unlike the Monte Carlo method employed in Chapter 8, the algorithm described here does not involve a temperature parameter to enable changes that increase the energy. However, it

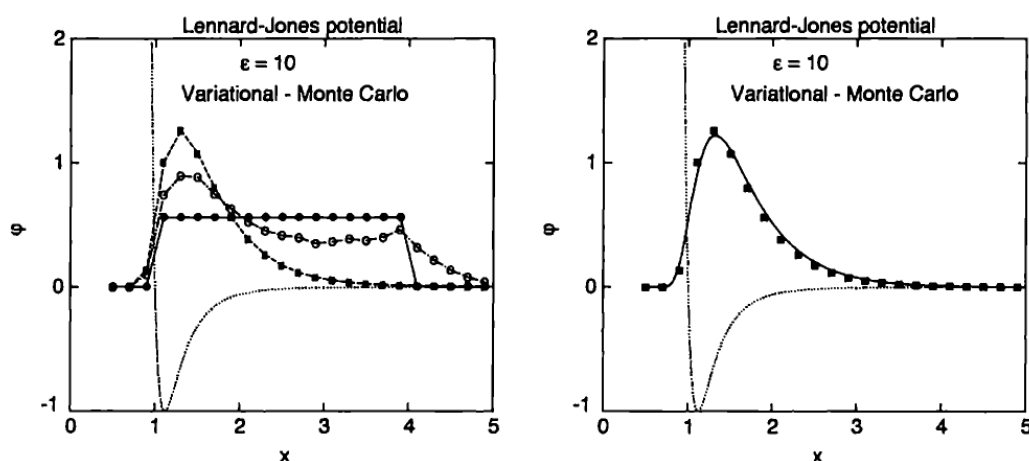


FIGURE 10.14: Left trial wave functions, φ , during the course of a variational–Monte Carlo calculation for the Lennard–Jones problem. The filled circles show the initial trial function, φ^0 , the open circles show the trial function after 10^3 attempted Monte Carlo moves (i.e., changes in the wave function), and the squares show it after 10^4 attempted moves. E^* for these trial wave functions was 0.11, -1.43 , and -2.18 , respectively. Right comparison of the final variational wave function with the results of the matching method (Figure 10.8). The filled squares here are the best trial functions from the plot on the left. Note that both of these wave functions have been properly normalized and that the potential (the dotted curves) is not shown to scale.

We will leave the construction of a variational–Monte Carlo program to the exercises (a full listing of such a program is given at this book’s Web site). All of the key ingredients of such a program have been described above. We will mention only one programming subtlety here. You must decide at the outset whether or not the trial functions are to be properly normalized (and if so, when to do it). This arises when evaluating E^* , since if φ is normalized the denominator in (10.28) is unity and thus need not be computed. For this reason it is convenient to always normalize φ . However, if you perform this normalization after each tentative Monte Carlo modification of the trial function, you must make sure that you can return to the previous trial function if the Monte Carlo modification is rejected.

As the Monte Carlo process is repeated many times, both the trial function and the corresponding value of E^* should converge to those of the true ground state. Some results for the trial function for the Lennard–Jones problem at various stages of the calculation are shown on the left in Figure 10.14. After a relatively small number of Monte Carlo iterations (10^3 in Figure 10.14), φ exhibits fluctuations that arise because this function was constructed from a small collection of random changes. However, after a large number of Monte Carlo adjustments φ becomes a smooth function of x , as we would expect for the true wave function. The final trial

is often useful to allow such energy increases with some probability *temporarily*, reducing such probability gradually as the procedure goes forward and the energy approaches the ground state energy. This is called *simulated annealing* and is described in Appendix B. Strictly prohibiting energy increases would correspond to quenching the system instantly to zero temperature, while simulated annealing corresponds to a more gradual decrease in temperature.

function shown here had an energy $E^* = -2.18$, which agrees with the value found with the matching method (Figure 10.8) to within about 10 percent. The trial function itself also agrees well with the wave function calculated with the matching method, as shown on the right in Figure 10.14. There are some slight differences and we will show in the exercises that they are due to the finite grid size used in describing the variational wave function.

The variational-Monte Carlo method can thus deal effectively with one dimensional situations such as the Lennard-Jones problem. However, its real strength is that it can be applied with very little additional effort to treat two- and three-dimensional cases. To illustrate this we now consider a particle confined by a two-dimensional harmonic potential. The Schrödinger equation in this case is (10.1) with the potential

$$V(x, y) = \frac{1}{2} k_x x^2 + \frac{1}{2} k_y y^2, \quad (10.29)$$

where k_x and k_y are related to the curvatures of the potential in the x and y directions. The astute reader will notice that this problem can be separated into two independent one-dimensional harmonic oscillators, one along x and one along y , which can be solved analytically. We will use this exact solution to evaluate the usefulness of the variational-Monte Carlo method. Several problems for which there are no known exact analytic solutions will be considered in the exercises.

To implement the variational approach for the two-dimensional oscillator we discretize space along the x and y directions into segments of length Δx and Δy , and write our trial functions as $\varphi(m, n)$, with $x = m\Delta x$, and $y = n\Delta x$ (for convenience we take $\Delta y = \Delta x$). We also restrict x and y to the range $-N\Delta x \leq x \leq N\Delta x$ and $-N\Delta x \leq y \leq N\Delta x$, so that we have $(2N+1)^2$ bins altogether. For this calculation we again pick an extremely simple initial trial wave function, $\varphi^0 = \text{constant}$ for all points in the region $|x| \leq 1.6$ and $|y| \leq 1.6$, and zero otherwise. This trial function is shown in Figure 10.15.

The Monte Carlo algorithm can then be implemented as in our previous example. During each Monte Carlo step a grid location ($m_{\text{test}}, n_{\text{test}}$) is chosen at random, and a provisional function obtained by changing the initial trial function $\varphi^0(m_{\text{test}}, n_{\text{test}})$ by a small amount chosen randomly from the range $\pm d\varphi$. If E^* for this provisional function is lower than that of φ^0 , then the provisional function becomes the new trial wave function. If E^* is higher, we keep φ^0 as the trial function. Some results are given in Figure 10.15, where on the left we show the trial function as the calculation proceeds. The final trial function has a roughly Gaussian shape, which may be familiar to you from the analytic solution of the harmonic oscillator. Figure 10.15 also shows the behavior of E^* during the calculation. It is a monotonically decreasing function, as required by the Monte Carlo procedure. After many Monte Carlo time steps it approaches a constant, which signals the convergence of the algorithm.

A plot of the final trial wave function as a function of both x and y is shown in Figure 10.16. The wave function falls off more slowly in the x direction than along y , since in this example we have chosen $k_x < k_y$. This means that the “spring constant” for vibrations along x was softer than the one along y , so the effective amplitude of vibration was larger along x . The value of E^* for this trial function

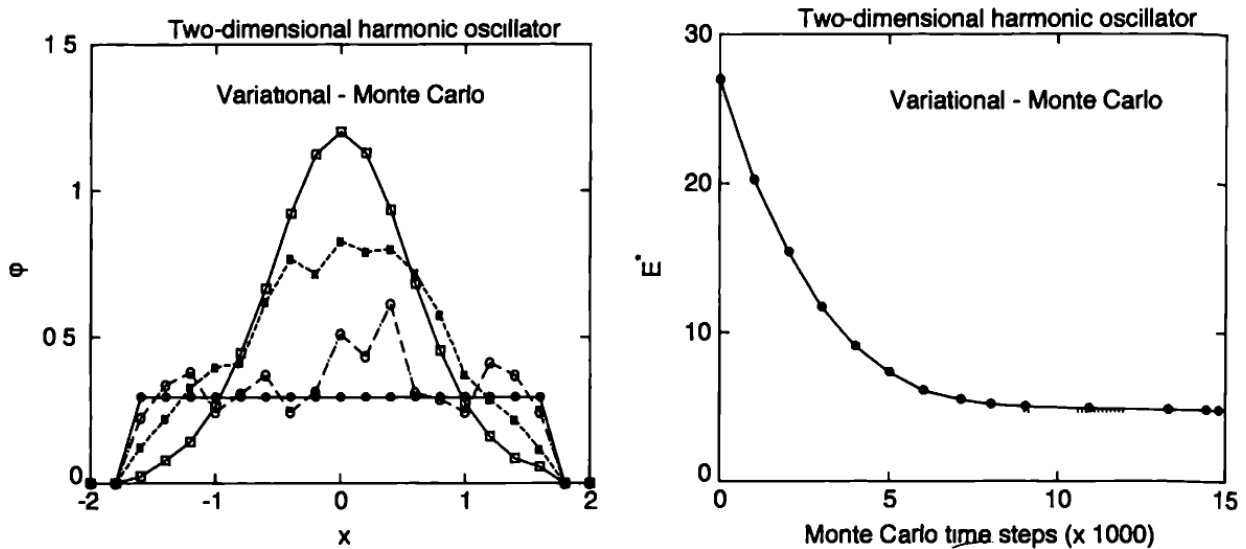


FIGURE 10.15: Left trial wave functions plotted as $\phi(x,0)$ as a function of x , at various stages in the variational–Monte Carlo procedure. The filled circles show the initial trial function, and the open circles, filled squares, and open squares show ϕ after 2000, 5000, and 15,000 Monte Carlo time steps. Right variation of E^* as the calculation proceeds. The potential energy was (10.29) with $k_x = 10$ and $k_y = 40$, the maximum values of $|x|$ and $|y|$ were 2.0 (so that $\phi = 0$ outside this region), and the grid size was $\Delta x = \Delta y = 0.2$.

is ~ 4.79 , which differs from the exact value $4.743\dots$ by about 1 percent. We will see in the exercises that this small difference is due to the finite grid size.

While the specific variational–Monte Carlo approach we have described here is fairly well suited for dealing with two- and three-dimensional problems, it is worth considering how the method can be improved. One difficulty is connected with the number of elements of the trial wave function $\phi_{m,n}$. For the two-dimensional harmonic oscillator example considered above there were ~ 400 grid elements. In order for the Monte Carlo strategy to be successful, it is necessary for each of these $\phi(m,n)$ to be adjusted many times, and if the number of grid elements is large this can require a long computation. This difficulty can be dealt with in several ways.

First, it pays to make a good choice of the initial trial function, as then an accurate estimate for the true wave function can be reached with fewer Monte Carlo iterations. Here “good” means that ϕ^0 should have a form close to the true ground-state wave function. We didn’t attempt to do this in our calculations here, but it is usually possible to make an informed choice for ϕ^0 . In the exercises we will explore how this can speed up convergence.

Second, in our oscillator example we chose the ranges of x and y to be large enough that the boundaries were far from the region where the wave function was nonzero. This was by design, since ideally the boundary condition $\phi = 0$ should be applied at $x, y = \pm\infty$. However, this means that there will be many grid elements where ϕ is extremely small and it would be nice to arrange for the Monte Carlo algorithm to spend less time updating $\phi(m,n)$ in these regions, since they

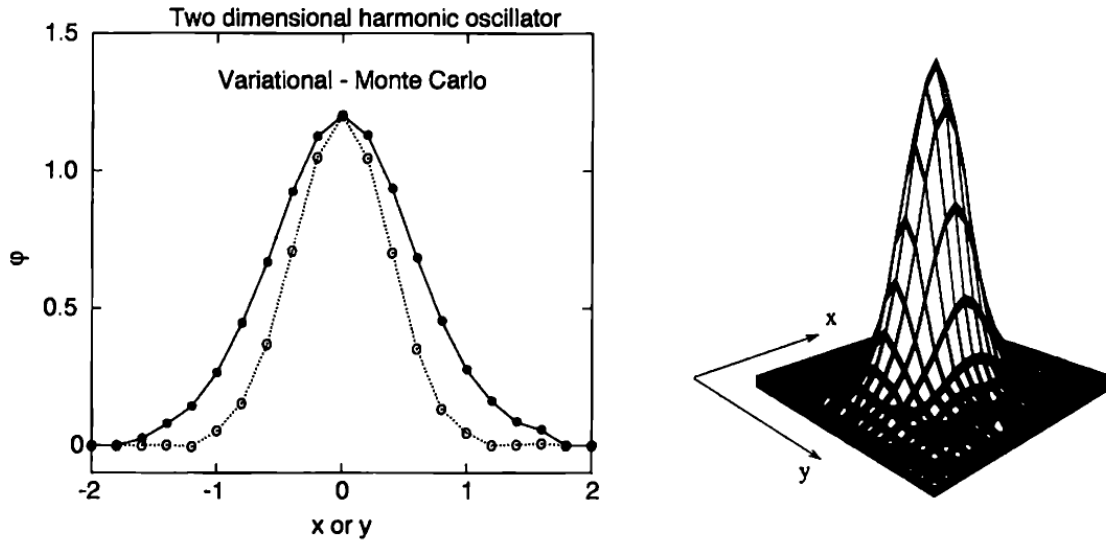


FIGURE 10.16: Left: plot of $\varphi(x,0)$ as a function of x (filled circles) and $\varphi(0,y)$ as a function of y (open circles) for the final trial wave function from Figure 10.15. Right: three-dimensional plot of the wave function.

are not as important as the regions where the wave function is large. This can be accomplished by modifying the way the test locations for the Monte Carlo changes are chosen. In our discussion above we chose the grid element to be considered in the variational procedure, m_{test} and n_{test} , at random from the *entire* allowed range of m and n . It would be better to choose the test point with a probability proportional to the value of φ at that point, since then the points where φ is largest would get the most attention from the algorithm. To do this requires that we be able to generate random numbers with a specified distribution. This is a problem that is discussed in Appendix F, where we consider two ways to generate random numbers with a nonuniform distribution. In the problem we are now considering the rejection method is most useful since we do not have an analytic expression for the desired probability function. The method is described in Appendix F, so we will not go into much detail here. A routine that generates m_{test} and n_{test} with a probability proportional to $\varphi(m,n)$ is sketched below.

EXAMPLE 10.2 Routine `calc_nxy` to generate random integers according to a specified weighting function

- Assume $\varphi_{\text{max}} \equiv \varphi(0,0)$ is the largest value of the trial function. This value is used to normalize $\varphi(x,y)$.
- Generate a point (x,y) randomly.
 - ▷ Generate two uniform random numbers r_1 and r_2 between 0 and 1

- ▷ Convert r_1 to a random integer x in the range $(-N, N)$ (use $x = \text{int}(r_1 \times (2N + 1)) - N$ where *int* means truncation to the integer part).
- ▷ Do the same for r_2 to get a random integer y
- Generate a third uniform random number r_3 and compare it with the normalized $\varphi(x, y)$. If $r_3 > \varphi(x, y)/\varphi_{\max}$, reject this choice of (x, y) and try again with a new set of 3 random numbers. Otherwise, accept it and put it into an array of accepted points.
- Repeat this procedure as many times as we need such points.

This routine generates the values of *mtest* and *ntest* as follows. A uniform random number generator is first used to generate points that are uniformly distributed over the entire allowed region of x and y . Given these uniformly distributed points we then *accept* each with a probability proportional to $\varphi(x, y)$ (otherwise they are rejected). The routine `calc_nxy` can readily be used in our variational-Monte Carlo calculations. In fact, we already used it in our work on the two-dimensional harmonic oscillator! In the exercises we will consider how much it speeds up the calculation as compared with simply choosing *mtest* and *ntest* uniformly.²⁷

A third useful modification of the variational-Monte Carlo method involves the manner in which we represent the trial functions. So far we have taken a direct, but in some ways crude approach by simply storing the value of φ at each grid site separately. A different strategy, which is in the spirit of a Fourier decomposition, would be to write the trial function as a sum of *basis* functions.²⁸ For example, with the harmonic oscillator we might choose the basis set to be Gaussian-like functions with different widths. The important point is that these functions be orthogonal so that an arbitrary trial function can be written uniquely in a form such as

$$\varphi(x, y) = \sum_k a_k b_k(x, y), \quad (10.30)$$

where $b_k(x, y)$ are the basis functions and the coefficients a_k specify the overlap between the trial wave function φ and the different basis functions. The variational procedure then involves adjusting the values of these coefficients so as to minimize the effective energy E^* . This can again be accomplished using a Monte Carlo approach.

EXERCISES

- 10.8. Write a program to implement the variational-Monte Carlo approach and apply it to the two-dimensional harmonic oscillator. Study how the choice of initial trial wave function affects the number of Monte Carlo steps required to obtain

²⁷In most cases it is advisable to initially choose *mtest* and *ntest* from a uniform distribution. This prevents errors (i.e., zeros) in the initial trial wave function from unduly restricting the form of later trial functions. After the trial function begins to resemble the true wave function, you can then switch to choosing from a biased distribution and thereby speed up the final convergence of the Monte Carlo algorithm.

²⁸We will encounter such basis functions in detail later in this chapter.

a “good” wave function. Here the “goodness” of a wave function can be judged from E^* . Compare the results for the very crude φ^0 considered in the text with that found for a Gaussian function. Other things to study include the influence of grid size on φ and E^* and the speed of convergence with and without a biased choice of the test points. *Hint:* Convergence can be determined by requiring that the change in E^* per iteration be less than some specified value. However, since some Monte Carlo iterations will not alter φ (and hence not change E^*), the convergence test cannot be performed after every iteration. Rather, you should make the test after every 10 or 100 iterations. A plot of E^* as a function of Monte Carlo time is also useful in testing for convergence.

- 10.9. Employ the variational–Monte Carlo approach to obtain the ground-state wave function and energy for the one-dimensional square well. Compare your results with those obtained in Section 10.2 with the shooting method.
- *10.10. Perform a variational–Monte Carlo calculation using an approach based on expressing the trial wave functions as functions with the form (10.30). Use the potential of your choice (it might be best to start with a simple square well).
- 10.11. Use the variational–Monte Carlo method to find the ground-state wave function for a particle in a box in three dimensions. Compare the results for boxes with different wall potentials.
- 10.12. Employ the variational–Monte Carlo method to calculate the ground-state energy and wave function of the anharmonic oscillator whose potential is given by $V(x) = x^4$.
- *10.13. The variational theorem as stated at the beginning of this section concerns the ground-state energy and wave function. That is why all of the problems in this section have (so far) been concerned with the ground state. However, the method can also be used to deal with the excited states in the following way. The wave functions of the ground and excited states form a complete orthogonal set of basis functions. If the variational function φ is orthogonal to the ground-state wave function, then the final result of a variation calculation will be the wave function of the first excited state and E^* will be the energy of this state. While this may sound simple in principle, a difficulty in practice is that even though φ^0 may be constructed so as to be orthogonal to the ground state, the Monte Carlo changes may spoil this orthogonality.
Investigate this problem by using the variational–Monte Carlo method to estimate the first excited state of a particle in a square well. The challenging part of the problem is to ensure in some manner that after each Monte Carlo modification of φ , this function is still orthogonal to the ground-state wave function.

10.5 TIME-DEPENDENT SCHRÖDINGER EQUATION: DIRECT SOLUTIONS

In the past few sections we have explored several approaches for solving the *time-independent* Schrödinger equation. We have seen how to calculate the wave functions of the ground and excited states for different potentials in one, two, and three dimensions. However, to appreciate some of the most interesting aspects of quantum mechanics it is necessary to consider the time dependence of the wave function. This will be our aim in the remainder of this chapter. Our first goal is to treat problems such as wave-packet propagation in free space, reflection from a potential step, and tunneling through a barrier.

The time-dependent Schrödinger equation has the form

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r}) \psi = i\hbar \frac{\partial \psi}{\partial t}, \quad (10.31)$$

where t is the time and again $i \equiv \sqrt{-1}$. Dealing with this problem in three dimensions is a formidable computational problem, so we consider the (numerically) simpler case of one dimension. We then have

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi = i\hbar \frac{\partial \psi}{\partial t}, \quad (10.32)$$

where now ψ is a function of x and t . We will explore a number of ways of dealing with this problem. One is a spectral method, much like the one we employed in connection with waves on a string in Chapter 6; we will describe that approach in Section 10.7. Here and in the next section (10.6) we consider somewhat more direct attacks.

The first and main approach we consider is a rather straightforward extension of the centered difference method used in Chapter 6. It is instructive to rewrite (10.32) in the form

$$i\hbar \frac{\partial \psi}{\partial t} = \mathcal{H} \psi, \quad (10.33)$$

where the Hamiltonian operator is defined by

$$\mathcal{H} \equiv -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x). \quad (10.34)$$

Constructing finite difference equations from these relations requires that we deal with complex quantities.²⁹ Thus, let us denote the real part of ψ by R and the imaginary part by I ,

$$\psi(x, t) = R(x, t) + i I(x, t). \quad (10.35)$$

Equation (10.33) then breaks into

$$\frac{\partial R}{\partial t} = \left(-\frac{1}{2} \frac{\partial^2}{\partial x^2} + V(x) \right) I \quad (10.36)$$

$$\frac{\partial I}{\partial t} = \left(\frac{1}{2} \frac{\partial^2}{\partial x^2} - V(x) \right) R, \quad (10.37)$$

where the factors involving $\hbar$ and m have (according to common convention for such problems) both been set equal to unity.³⁰ We then approximate both the spatial and the time derivatives as centered differences. For example,

$$\frac{\partial^2 R(x, t)}{\partial x^2} \approx \frac{R(x + \Delta x, t) - 2R(x, t) + R(x - \Delta x, t)}{(\Delta x)^2}, \quad (10.38)$$

²⁹Unlike the time-independent problems considered in the previous sections, for most time-dependent problems we can not avoid a complex wave function.

³⁰This is a slightly different convention than what we used in discussing the time-independent Schrödinger equation in earlier sections.

and

$$\frac{\partial I(x, t)}{\partial t} \approx \frac{I(x, t + \Delta t/2) - I(x, t - \Delta t/2)}{\Delta t} . \quad (10.39)$$

Inserting these into (10.37) yields an expression for $I(x, t + \Delta t/2)$ in terms of $I(x, t - \Delta t/2)$, $R(x \pm \Delta x, t)$, and $R(x, t)$. After a little rearranging we get

$$\begin{aligned} I(x, t + \Delta t/2) &\approx I(x, t - \Delta t/2) \\ &+ \frac{\Delta t}{2(\Delta x)^2} [R(x + \Delta x, t) - 2R(x, t) + R(x - \Delta x, t)] \\ &- (\Delta t)V(x)R(x, t) . \end{aligned} \quad (10.40)$$

If we now discretize space by taking $x_m = m\Delta x$, and use the values of R and I at x_m , it is evidently most convenient, according to (10.40), to evaluate $R(x, t)$ at $x = x_m$ and $t = n\Delta t$ while $I(x, t)$ is evaluated at $x = x_m$ and at the *shifted* times $t = (n \pm \frac{1}{2})\Delta t$. The corresponding expression for the real part of the wave function R using the shifted discrete times for R and I is

$$\begin{aligned} R(x, t + \Delta t) &\approx R(x, t) - \frac{\Delta t}{2(\Delta x)^2} [I(x + \Delta x, t + \Delta t/2) - 2I(x, t + \Delta t/2) \\ &+ I(x - \Delta x, t + \Delta t/2)] - (\Delta t)V(x)I(x, t + \Delta t/2) . \end{aligned} \quad (10.41)$$

We are thus evaluating the real and imaginary parts of $\psi(x, t)$ in a *leapfrog* fashion at “staggered” times. However, this presents no practical difficulty, and in fact one can show that the numerical errors made in this method are locally of order $(\Delta t)^3$, the same order as with the second-order Runge-Kutta methods. This leapfrog approach also has the very important property of conserving the normalization of the wave function; that is, the integral of $\psi^*\psi$ over all space is conserved.

While the leapfrog method just described is suitable for carrying out the calculations to obtain all of the results that we show later in this section, we will now describe a second direct approach which is related to the general matrix formulation that we discussed earlier in this chapter. This second method provides an important lesson in dealing with linear equations containing Hermitian operators such as $\mathcal{H}$. Although (10.33) has the appearance of a simple first-order differential equation involving the time dependence of ψ , things are complicated by the fact that $\mathcal{H}$ is an operator rather than a number. Ignoring that fact for a moment, (10.33) has the formal solution

$$\psi(x, t) = \exp(-i t \mathcal{H}/\hbar) \psi(x, 0) , \quad (10.42)$$

where $\psi(x, 0)$ is the wave function at $t = 0$. Thus, if we know $\psi(x, 0)$, we can *formally* calculate the behavior at all future times using (10.42). However, this formal solution is not as useful as we might hope because of the operator $\mathcal{H}$ in the exponent. To be clear on what it means mathematically to have an operator expression of this kind, it is useful to rewrite (10.42) using the Taylor expansion of the exponential factor

$$\exp(-i t \mathcal{H}/\hbar) \psi = \left(1 - \frac{i t}{\hbar} \mathcal{H} - \frac{t^2}{2 \hbar^2} \mathcal{H}^2 \dots \right) \psi , \quad (10.43)$$

where $\mathcal{H}^n$ means that the operator $\mathcal{H}$ is applied n times in succession. The problem comes when we have to evaluate terms such as $\mathcal{H}^2\psi$. Recalling the definition of $\mathcal{H}$, we see that such a term expands to give

$$\mathcal{H}^2\psi = \mathcal{H}\mathcal{H}\psi = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right] \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right] \psi. \quad (10.44)$$

Difficulties now arise from the fact that the two terms that make up $\mathcal{H}$ do not commute; that is, the order of operation is important. We cannot simply switch the order with which we apply the derivative term in $\mathcal{H}$ (the kinetic energy piece) and the potential energy part (which, in general, depends on position). This makes dealing with the time-dependent Schrödinger equation more complicated than dealing with a simple first-order differential equation such as those discussed in Chapter 1.

While (10.42) is a perfectly fine formal solution, it is not of much practical use since the expansion (10.43) involves a very large (infinite) number of terms. However, this formal solution does suggest a way to proceed numerically. Let us consider applying the formal solution over a very small time interval. Indeed, let us discretize time in steps Δt ; using (10.42) we obtain

$$\psi(x, t + \Delta t) = \exp(-i \Delta t \mathcal{H}/\hbar) \psi(x, t). \quad (10.45)$$

But, you might ask, how can this be a useful approach in view of the difficulties with the expansion (10.43)? The reason we could have some optimism with regard to (10.45) is that a small value of Δt will make the higher-order terms in the expansion (10.43) small. Let us assume for now that these terms can be made small enough that we can stop at the term linear in $\mathcal{H}$, which yields³¹

$$\psi(x, t + \Delta t) \approx (1 - i \Delta t \mathcal{H}/\hbar) \psi(x, t) = \left[1 + \frac{i \Delta t}{2} \frac{\partial^2}{\partial x^2} - i \Delta t V(x)\right] \psi(x, t), \quad (10.46)$$

where in the last step we have again assumed $\hbar = 1$ and $m = 1$. Given (10.46), a natural way to proceed is to discretize space into units of size Δx and write the wave function as $\psi(m, n) \equiv \psi(m\Delta x, n\Delta t)$. Expressing the second partial derivative in the usual finite-difference form, (10.46) becomes

$$\begin{aligned} \psi(m, n+1) \approx \psi(m, n) + \frac{i \Delta t}{2} \left[\frac{\psi(m+1, n) + \psi(m-1, n) - 2\psi(m, n)}{(\Delta x)^2} \right] \\ - i \Delta t V(m) \psi(m, n). \end{aligned} \quad (10.47)$$

Hence, given $\psi(m, n)$ it appears that we can readily calculate $\psi(m, n+1)$, the wave function at the next time step. However, this approach contains two flaws.

The first flaw is this procedure is numerically unstable! The effects of small round-off and other types of errors (due, for example to the higher-order terms we have neglected) grow rapidly with time, essentially because of the exponential factor in (10.45) from which this finite-difference equation was derived. Amazingly,

³¹This could also be obtained by expressing (10.33) in finite-difference form, but we will have further use for (10.45) moment

this instability can be cured by rewriting (10.45) so as to, in a sense, propagate *backward* in time,

$$\psi(x, t) = \exp(+i \Delta t \mathcal{H}/\hbar) \psi(x, t + \Delta t) \approx (1 + i \Delta t \mathcal{H}/\hbar) \psi(x, t + \Delta t). \quad (10.48)$$

We will not go into the details of precisely why the scheme (10.48) leads to numerical stability, but leave that for interested readers to explore using the references. However, this stability does come at a price; when (10.48) is converted into finite difference form one is led to the following set of the finite-difference equations

$$\begin{aligned} \psi(m, n) \approx \psi(m, n + 1) & \quad (10.49) \\ - \frac{i \Delta t}{2} \left[\frac{\psi(m + 1, n + 1) + \psi(m - 1, n + 1) - 2 \psi(m, n + 1)}{(\Delta x)^2} \right] \\ + i \Delta t V(m) \psi(m, n + 1). \end{aligned}$$

Comparing this with (10.47) we find that it is an *implicit* relation for $\psi(m, n + 1)$. That is, we cannot use (10.49) to express $\psi(m, n + 1)$ solely in terms of the wave function at time step $n + 1$, but we are stuck with terms of the form $\psi(m', n + 1)$ with $m' \neq m$ in the expression for $\psi(m, n + 1)$. Hence, we cannot just step through the values of m and use (10.49) to calculate the wave function at the next time step. We must instead solve a system of algebraic equations, and this amounts to a matrix formulation as described earlier in this chapter.

However, even this price is not all that we must pay. Equations (10.47) and (10.49) both violate a fundamental property of quantum mechanics called *unitarity*. The Schrödinger equation preserves the proper normalization of a wave function, that is, the total probability for finding a particle somewhere in space must always be unity.³² It turns out that $(1 - i \Delta t \mathcal{H}/\hbar)$, the time evolution operator in (10.46), is *not* unitary, and the same is true of (10.48). We must therefore devise a different strategy.

A numerically stable approach that does satisfy unitarity involves writing the exponential factor in (10.45) in what is known as the Cayley form

$$\exp(-i \Delta t \mathcal{H}/\hbar) \approx \frac{1 - i \Delta t \mathcal{H}/2\hbar}{1 + i \Delta t \mathcal{H}/2\hbar}. \quad (10.50)$$

We will leave it to the reader to show that to lowest order in $\mathcal{H}$ (in a Taylor expansion), this is equal to both $1 - i \Delta t \mathcal{H}$ and $(1 + i \Delta t \mathcal{H})^{-1}$, which are the key factors in (10.46) and (10.48). However, these three ways of approximating the exponential factor $\exp(-i \Delta t \mathcal{H})$ are *not* equivalent with regard to maintaining unitarity, as can be seen in the following way. Using (10.50) to propagate our wave function forward in time we have

$$\psi(x, t + \Delta t) \approx \frac{1 - i \Delta t \mathcal{H}/2\hbar}{1 + i \Delta t \mathcal{H}/2\hbar} \psi(x, t). \quad (10.51)$$

³²The conservation of probability is ensured by the *unitarity* of the operator $\exp(-i t \mathcal{H}/\hbar)$ in (10.42), which is the direct result of the fact that $\mathcal{H}$ is Hermitian.

From (10.51) we first obtain

$$\left[1 + \frac{i \Delta t \mathcal{H}}{2\hbar}\right] \psi(x, t + \Delta t) = \left[1 - \frac{i \Delta t \mathcal{H}}{2\hbar}\right] \psi(x, t) \quad (10.52)$$

We now replace $\mathcal{H}$ by (10.34), convert everything to finite-difference form, and discretize space and time so that $\psi(m, n) \equiv \psi(m\Delta x, n\Delta t)$. After rearranging a few terms we get

$$\begin{aligned} \psi(m+1, n+1) + [2i\lambda - 2(\Delta x)^2 V(m) - 2] \psi(m, n+1) + \psi(m-1, n+1) \\ = -\psi(m+1, n) + [2i\lambda + 2(\Delta x)^2 V(m) + 2] \psi(m, n) - \psi(m-1, n), \end{aligned} \quad (10.53)$$

where for notational convenience we define $\lambda \equiv 2(\Delta x)^2/\Delta t$. The system of equations (10.53) has a form that is different from anything we have encountered in previous chapters. Most importantly, they cannot be rewritten so as to express $\psi(m, n+1)$ solely in terms of $\psi(m, n)$. Hence, we cannot simply loop through the values of m and compute each $\psi(m, n+1)$ in a straightforward manner. Instead, we have an *implicit* numerical problem, which means that the value of $\psi(m, n+1)$ for a particular value of m depends on $\psi(m', n+1)$, where $m' \neq m$. That is, we have a set of algebraic equations (which can again be cast as a matrix problem) that must be solved to obtain the wave function at the next time step.

As noted earlier in this chapter, many eigenvalue problems can be expressed in matrix form, with the solution then requiring that a matrix be diagonalized or inverted. In the present case we must invert the matrix whose coefficients can be read from (10.53). The bad news is that this will be a *large* matrix. In the particular problems we will consider below, the number of grid elements along x is typically 1000, so the associated matrix is 1000×1000 in size. This is a formidable computational problem, especially when we realize that such a matrix must be inverted for *each* time step! Fortunately, there is some good news: most of the elements of this matrix are zero. If we examine (10.53) we see that the equation for $\psi(m, n+1)$ involves just $\psi(m \pm 1, n+1)$, and no other values of ψ [except for itself, $\psi(m, n+1)$]. Thus, each row of our matrix has only three nonzero elements. Since one of these is on the diagonal and the other two are just one space away, this matrix is *tridiagonal*. Such matrices arise often in numerical work, since as in the present case the effectively “local” nature of the underlying differential equation gives rise naturally to this matrix structure. As you might expect, the large number of the zeros in the matrix can be used to our advantage. There are several algorithms which take advantage of the special form and give good results. A popular approach is the *Crank-Nicholson* method, which is described in detail in the paper by Goldberg, Schey, and Schwartz in the references (this method was also discussed in detail in the first edition of this book). The Crank-Nicholson approach and the leapfrog method described earlier in this section can both be used for the calculations we describe in the remainder of this section.

We now consider a few examples of the time-dependent behavior of ψ as calculated using one of the algorithms described above. The primary initial condition

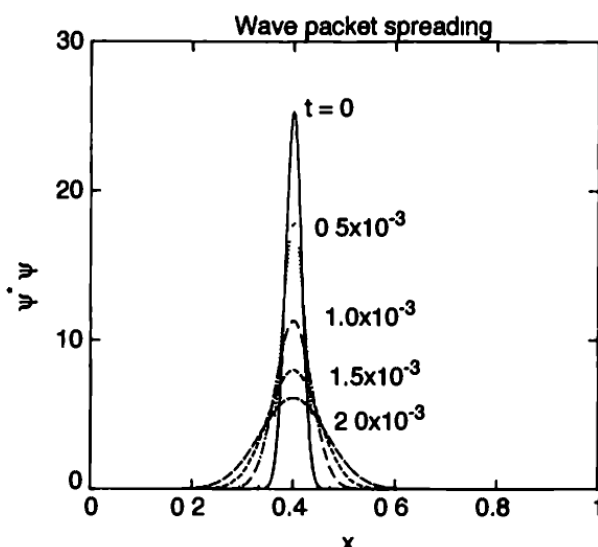


FIGURE 10.17: Time-dependent behavior of a Gaussian wave packet. The initial wave packet ($t = 0$) is given by (10.54) with $x_0 = 0.4$ and $\sigma^2 = 1 \times 10^{-3}$. The wave function at later times (the time values are given in the figure) is seen to spread out in space. For this calculation we used $\Delta x = 0.0005$ with the boundary condition $\psi = 0$ applied at $x = 0$ and $x = 1$. We also took $\Delta t = 5 \times 10^{-7}$ so that $\lambda = 2(\Delta x)^2/\Delta t = 1$.

in these calculations is the initial wave function. In all of the examples considered below we assume a Gaussian form for ψ at $t = 0$, so that our initial wave function is given by

$$\psi(x, t = 0) = C \exp[-(x - x_0)^2/\sigma^2], \quad (10.54)$$

where the center of the packet is at x_0 , its width is σ , and the factor C is chosen to satisfy the normalization condition on ψ . A plot of such a wave packet is shown by the solid curve in Figure 10.17. This wave function is not stable with time; we see from Figure 10.17 that it broadens as t increases. Since our algorithm maintains unitarity³³ a broader wave function must have a lower value at the peak, as also seen in Figure 10.17.

The physics responsible for this wave-packet spreading can be understood by recalling that we are dealing with an equation much like the wave equation considered in Chapter 6. The Gaussian wave packet (10.54) can be decomposed into (i.e., written as) a sum of plane waves of the form e^{ikx} . These plane waves are essentially just Fourier components³⁴ whose amplitudes are chosen so that they add up to give the Gaussian packet (10.54) at $t = 0$. However, each component is a wave that travels along the system. These waves have different wave vectors, k ,

³³It is a good idea to periodically check that $\int \psi^* \psi dx = 1$, as this provides a strong test that your program does not contain any subtle bugs. The wave functions shown in Figure 10.17 and in later figures all satisfy this test. With the leapfrog algorithm, we need to sum the squares of the real and imaginary parts evaluated half a time step apart to obtain ψ^2 , but this usually presents no practical problem.

³⁴Just as the sines and cosine standing waves are the Fourier components for waves on a string

and thus different velocities. In addition, half of them travel in the $+x$ direction and half along $-x$. The motion of the components causes the wave packet to spread with time. This can also be interpreted using the Heisenberg uncertainty principle, according to which $\delta x \delta p \geq \hbar/2$, where δx and δp are the widths of the wave packet in terms of space and momentum. Since the momentum $p = \hbar k$, the uncertainty relation tells us that confining the particle to an initial wave packet of width δx (which in Figure 10.17 is ~ 0.05) requires an uncertainty in k , that is, a corresponding spread in the values of k , and this makes the packet spread out with time.

The simple packet (10.54) includes components with both positive and negative wave vectors in equal amounts. This makes it a stationary packet in the sense that the average position $\bar{x} \equiv \int \psi^* x \psi dx$ does not change with time. This is not the case for the packet

$$\psi(x, t = 0) = C \exp[-(x - x_0)^2/\sigma^2] \exp[i k_0 x] \quad (10.55)$$

which is just our original Gaussian function multiplied by a plane wave with wave vector k_0 . This wave function, which is complex, can also be written as the sum of components with a range of k values, but with the range now centered around k_0 . As a result, this packet travels with an average velocity $v_0 = \hbar k_0/m$. It is interesting to compare it with the stationary wave functions considered above. While we only show $\psi^* \psi$ in Figure 10.17, you can see from (10.54) that at $t = 0$ this is also just the square of the real part of ψ and that the imaginary part is zero. Our propagating packet is a little more complicated, as can be seen from Figure 10.18. While $\psi^* \psi$ has the same simple Gaussian shape, the real and imaginary parts of ψ oscillate rapidly. This oscillation is due to the factor $\exp(ik_0 x)$ in (10.55). In fact, for these oscillations to be prominent, we must choose k_0 and σ appropriately; i.e., we need $\sigma \gg 1/k_0$. This choice also makes $\hbar k_0$ a well-defined mean momentum of the packet, as the width of the wave packet in k -space is proportional to $1/\sigma$.

In carrying out the calculations, we should also not forget the stability and accuracy requirements. As in any numerical solution of initial-value problems, the calculated wave function can propagate at a speed that is no faster than $\Delta x/\Delta t$, where Δx and Δt are the discrete steps in space and time, respectively. Since the wave packet moves on average with a speed $v_0 = \hbar k_0/m$, we must have $\Delta x/\Delta t \geq \hbar k_0/m$ for the calculation to keep up with the packet. Also an accurate description of the oscillating real and imaginary parts in the packet requires $\Delta x \ll \lambda_0$ where $\lambda_0 = 2\pi/k_0$ is the average wavelength.

The results of our propagation program when applied to this packet are shown in Figure 10.19. Here the packet in Figure 10.18 was allowed to move in a region with $V = 0$. That is, we had a freely moving particle. We will leave it to the exercises to show that this motion has the expected velocity. You should notice that the height of the propagating packet decreased as it moved. This is due to the wave-packet spreading we noted in connection with the stationary packet in Figure 10.17. The uncertainty relation tells us that *all* such packets will spread with time. If we had chosen a larger value of k_0 , the velocity of the packet would be larger, and since the rate of spreading would be the same, the packet would better

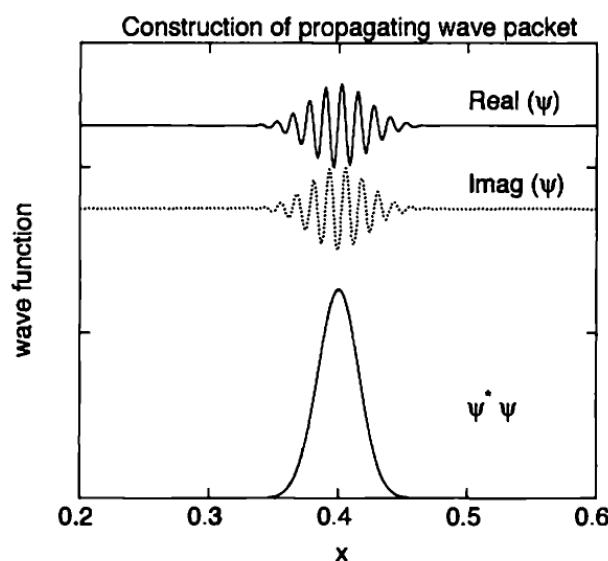


FIGURE 10.18: Composition of a propagating wave packet. For this packet we took $k_0 = 500$, $x_0 = 0.4$, and $\sigma^2 = 0.001$.

maintain its amplitude for a given distance traveled. We will investigate this in the exercises.

We can use this propagating wave packet to study a wide variety of problems involving reflection, transmission, and tunneling. The programming is essentially identical to that required to treat motion in a flat ($V = 0$) potential region. The only difference is that we must employ the appropriate potential function $V(x)$ in the calculation. We first consider an encounter with a potential wall; that is, we let our wave packet begin in a region with $V = 0$ and propagate toward a region where V is very large. Figure 10.20 shows the behavior with a wall located at $x = 0.6$. The potential for $x \geq 0.6$ was 1×10^6 . This was much greater than the average energy of the wave packet, $E_{\text{packet}} = \hbar^2 k_0^2 / 2m$, which in this case was 1.25×10^5 . Hence, the particle did not have enough energy to classically propagate into the region $x \geq 0.6$ and was instead completely reflected.

In some respects the reflection process resembles the behavior of a wave pulse on a string when it encounters the end of the string. However, the details in the quantum case are somewhat different as can be seen from the detailed plots on the right in Figure 10.20. The rapid oscillations of $\psi^* \psi$ are especially striking and are often not appreciated from the usual analytic treatments of the reflection process.

Another interesting case is a potential cliff. We again consider a wave packet that, as in Figure 10.20, was incident from a region in which $V = 0$. Now, however, the potential *dropped* to $V = -10^6$ when $x \geq 0.6$. From Figure 10.21 we see that while some of the packet was transmitted, a comparable amount was also reflected. That is, there was a nonzero probability for the particle to be reflected from the cliff! This is a nice illustration of the wave nature of quantum mechanics, since such behavior would never be found in classical mechanics. In a sense we could

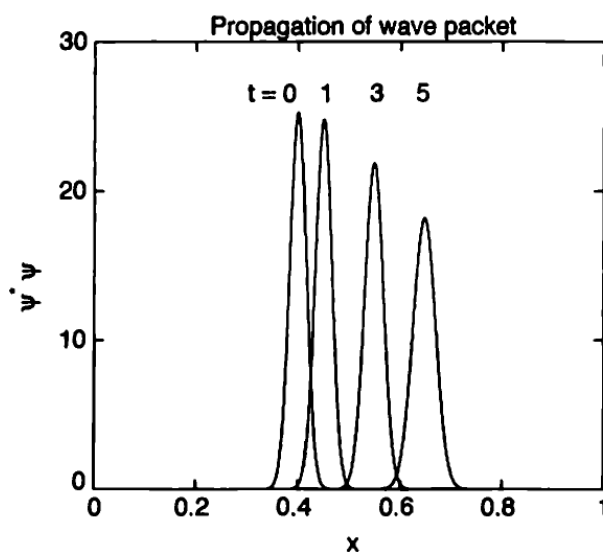


FIGURE 10.19: Wave packet (10.55) from Figure 10.18 propagating in a region with $V = 0$. The time values given in the figure are to be multiplied by $\times 10^{-4}$. We used $\Delta x = 0.0005$ and $\Delta t = 5 \times 10^{-7}$ in the calculation.

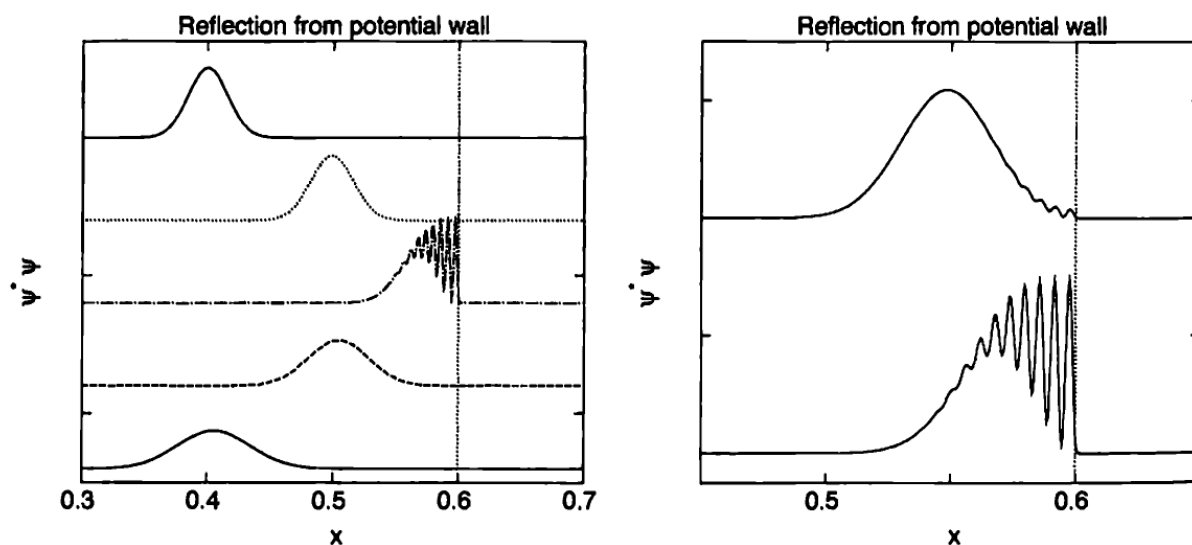


FIGURE 10.20: Wave packet reflecting from a wall, which is indicated by the vertical dotted lines. The wave-packet parameters were the same as those used in Figure 10.19. The potential was $V = 0$ on the left ($x < 0.6$) and $V = 1 \times 10^6$ on the right ($x \geq 0.6$). Left: plots of $\psi^* \psi$ at various times, with time increasing as we move down the sequence. The particle initially traveled from left to right, was reflected from the wall, and then traveled toward the left. Note that the wave packet at the bottom was significantly broader than the initial packet, due to the spreading effects discussed above. Right: expanded plot of two views recorded as the wave was "striking" the wall.

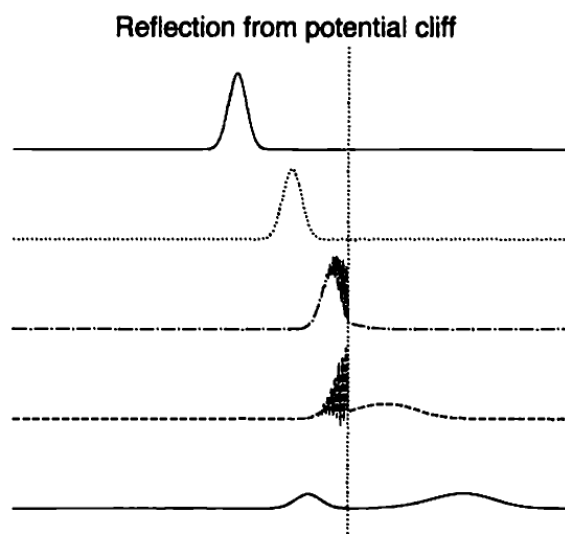


FIGURE 10.21: Wave packet incident on a cliff whose position is indicated by the vertical dotted line. The wave-packet parameters were the same as those used in Figure 10.20. The potential was $V = 0$ on the left ($x < 0.6$) and $V = -1 \times 10^6$ on the right ($x \geq 0.6$). These are plots of $\psi^*\psi$ at various times, with time increasing as we move down the graph. The particle initially traveled from left to right. After the encounter with the cliff, a small packet continued to the right, while another packet was reflected and traveled to the left. Note that the transmitted packet has a higher speed than the reflected one, since in this region the kinetic energy is higher (due to the lower potential energy).

conclude that a quantum world would be a safer place to live, since it would be harder to fall off a quantum cliff than a classical one.

As a final example in this section we consider a wave packet incident on a barrier. Here the potential was $V_0 = 1.25 \times 10^5$ in the region $0.6 \leq x \leq 0.7$, with $V = 0$ to the left and right of the barrier. We chose this value of V_0 since it was equal to the “average” kinetic energy for this wave packet, $\hbar^2 k_0^2 / 2m$. We see from Figure 10.22 that the incident packet was mainly reflected. If we had plotted the results on a greatly expanded scale, you would observe that a small amount was also transmitted. In addition, on the time scale considered here a small amount of the packet “remained behind” in the barrier region. The velocity in this region is essentially zero, since the average kinetic energy is $\hbar^2 k_0^2 / 2m - V = 0$, so it takes a long time for a wave packet in this region to escape.

The general approach to the time-dependent Schrödinger equation, which we have described in this section, can be applied to a very wide variety of problems. Further examples will be explored in the exercises.

EXERCISES

- 10.14.** Study the motion of a propagating wave packet in free space as in Figure 10.19 and show quantitatively that it does indeed move with a velocity of $\hbar k_0 / m$. Compare the behavior with different values of k_0 . You should find that as k_0 is made larger, our algorithm requires smaller spatial and time steps to avoid

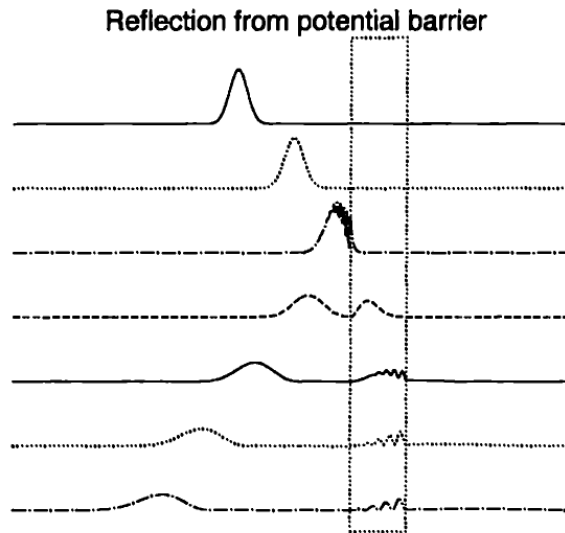


FIGURE 10.22: Wave packet partially reflecting from a potential barrier (indicated by the dotted lines) with $V = 1.25 \times 10^5$. The other parameters were the same as used in Figure 10.20. The packet initially traveled from left to right (top). Part of the packet was reflected and traveled away to the left, while a very small part (which is barely visible on the scale used here) was transmitted and proceeded to the right.

substantial numerical errors. These errors are not significant for the values of Δx and Δt we have used in the calculations in this section, but this would not have been the case if we had employed somewhat larger values of k_0 .

- 10.15.** Study how the rate of wave-packet spreading as observed in Figure 10.17 varies with the width of the initial packet. Compare this spreading quantitatively with the Heisenberg uncertainty relation. Do this by calculating the width of the wave packet as a function of time.
- 10.16.** Observe the propagation of a wave packet in a region in which the potential increases linearly with position. Study the reflection of the wave packet in this case.
- *10.17.** Study reflection and transmission from a barrier for which the potential is greater than the kinetic energy of the incident wave packet. The probability for the particle to tunnel through such a barrier decreases rapidly as the potential is increased. In order to have a significant transmission probability you will have to make the barrier thinner than in Figure 10.22. *Hint:* Good parameter choices are $k_0 = 700$, $\Delta t = 5 \times 10^{-7}$, and $\Delta x = 5 \times 10^{-4}$. Let the height of the potential barrier be $V_0 = 2 \times k_0^2$ with a width of 0.05. You should also explore the behavior for other values of both V_0 and the width of the barrier.
- *10.18.** Compute the scattering of a wave packet from the potential well shown in Figure 10.23. Try to observe effects associated with the bound states in the well. *Hint:* Start with the wave-packet parameters given in the previous problem. Assuming that the particle is incident from the left, take the height of the left-most barrier to be $4k_0^2$ and give it a width of 0.05. Give the right-most barrier the same width and a much larger height, and let the two barriers be separated by $L = 1$ (or less). You should find that the penetration of the wave function into

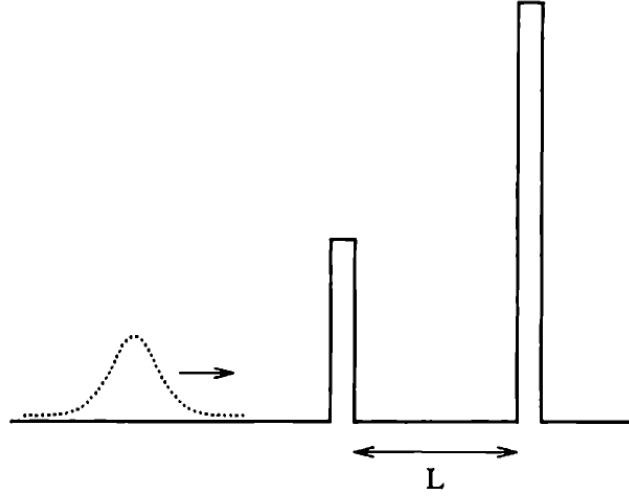


FIGURE 10.23: Schematic illustration of a wave packet (dotted curve) incident on a region containing two potential barriers $V = 0$ on the left, on the right, and between the barriers. The barrier heights are greater than the kinetic energy of the particle.

the potential well depends on the value of k_0 and that this penetration is greatest when the energy of the particle matches the energy of one of the approximate bound states of the well.

10.6 TIME-DEPENDENT SCHRÖDINGER EQUATION IN TWO DIMENSIONS

In this section we extend the leapfrog method introduced in the previous section to two dimensions. The further extension to three dimensions is straightforward, although plotting (and making sense of!) the results becomes more difficult as the spatial dimension grows.

When working in two dimensions, our discrete versions of the potential $V(x, y)$ and the functions $\psi(x, y, t) = R(x, y, t) + iI(x, y, t)$ have two spatial indices, and $\psi(x, y)$ must now be normalized on a two-dimensional grid. However, the modifications required to the numerical algorithm in (10.40) and (10.41) are relatively minor. The replacement of $\partial^2/\partial x^2$ in the one-dimensional Hamiltonian (10.34) by the two-dimensional Laplacian ∇^2 changes the spatial derivative term (10.38) for the real part of ψ (the function R), and for the imaginary part of ψ (the function I). We still use centered differences to approximate these derivatives, and for the real part of the wave function we get

$$\begin{aligned} \nabla^2 R(x, y, t) &\approx \frac{R(x + \Delta x, y, t) - 2R(x, y, t) + R(x - \Delta x, y, t)}{(\Delta x)^2} \\ &\quad + \frac{R(x, y + \Delta y, t) - 2R(x, y, t) + R(x, y - \Delta y, t)}{(\Delta y)^2} \\ &= \frac{R(x + \Delta x, y, t) + R(x - \Delta x, y, t) + R(x, y + \Delta y, t) + R(x, y - \Delta y, t) - 4R(x, y, t)}{(\Delta x)^2}, \end{aligned} \quad (10.56)$$

where we again set $\Delta x = \Delta y$ for simplicity. The spatial derivative term for the imaginary part of the wave function has a similar form. These can then be inserted into the two-dimensional versions of (10.36) and (10.37), and after some straightforward algebra one can obtain leapfrog relations for R and I which are analogous to (10.40) and (10.41). The incoming wave packet must also be two dimensional (by suitable generalization of (10.55)), with an initial wave vector of the form $\mathbf{k}_0 = k_{0x}\hat{x} + k_{0y}\hat{y}$. The constraints for numerical stability and accuracy as well as for the quantum effects are similar to those in one dimension.

The main computational challenge may actually be visualization. Unlike in one dimension, we need both the x and y directions to describe the position and thus we must find a way to represent the probability density $\psi^*\psi$. This is often done by plotting a perspective view of a three-dimensional surface where the z direction represents the probability density. However, another effective way is to use colors or shades of gray to represent $\psi^*\psi$ on a two-dimensional grid. In what follows, we will mostly use the latter method as it tends to require less computing and does not require sophisticated graphics routines.

Some typical results are shown in Figure 10.24, where the wave packet scatters from a cylindrical potential well of radius $a = 0.2$ centered at the origin of a square grid with x and y ranging from -1 to 1 . The well has a depth $V_0 = -1000$, and a Gaussian wave packet of incident energy $k_0 = 800$ is incident from the left. Successive plots describe the time evolution of the wave packet from $t = 0$ through $t = 0.02$ (the time step for the calculation was $\Delta t = 0.0001$). For this case of the kinetic energy of the incoming wave packet on the same order as the depth of the potential well, and we find large oscillations in the probability amplitude near the edges of the well (as we also observed in one-dimension), and most of the packet is eventually transmitted to the right. There is also a small reflected wave, but it is hard to see in this figure because its amplitude is very small. We also find that transmission and reflection occur not just along the incoming axis but also *around* the well, and the main transmitted amplitude spreads laterally as well. In two dimensions, there are simply many more paths the wave packet can take, allowing parts of the packet to avoid going *through* the well.

Another example is shown in Figure 10.25, where the same incoming wave packet impinges on a cylindrical potential barrier instead. The incoming wave packet energy is now a bit smaller than the barrier height, so one would expect large quantum effects. Recall that in one dimension this situation typically results in significant penetration of the packet into the classically forbidden barrier interior, followed by reflection and transmission, with a small third wave portion of the wave function remaining trapped inside the barrier. The result in two dimensions is somewhat different. From Figure 10.25, we see that the bulk of the incident wave packet goes *around* the barrier instead of *through* it. However, similar to one dimension, a small portion of the wave penetrates the barrier and lingers there for quite some time after the main part of the packet passes by, though this is hard to see from these figures. In order to illustrate the last point, we also show in Figure 10.26 the three-dimensional surface view at $t = 0.0250$. We clearly see a small peak right near the center of the barrier (the origin), while most of the wave packet has either gone around the barrier or been reflected.

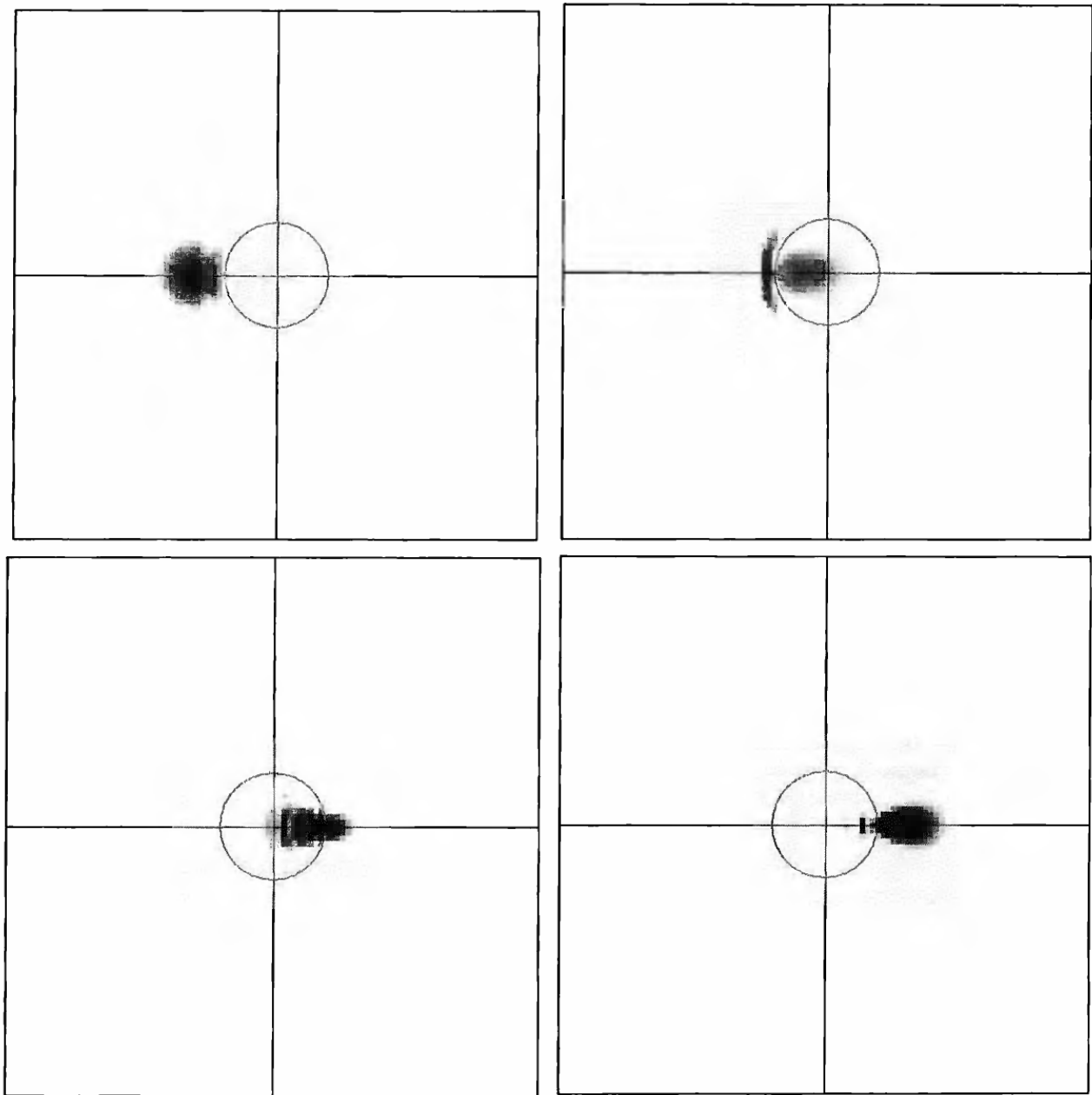


FIGURE 10.24: Wave packet of $k_0 = 40$ impinging on a well of depth $V_0 = -1000$, where $\sigma = 0.1$ and $x_0 = -0.5$. The shades of gray linearly progress toward black as the probability density rises. Shown on upper left is the probability density at $t = 50\Delta t = 0.0050$, upper right $t = 0.0100$, lower left $t = 0.0150$, and lower right $t = 0.0200$. The space is discretized in units of $\Delta x = 0.02$, x and y range between -1 and 1 .

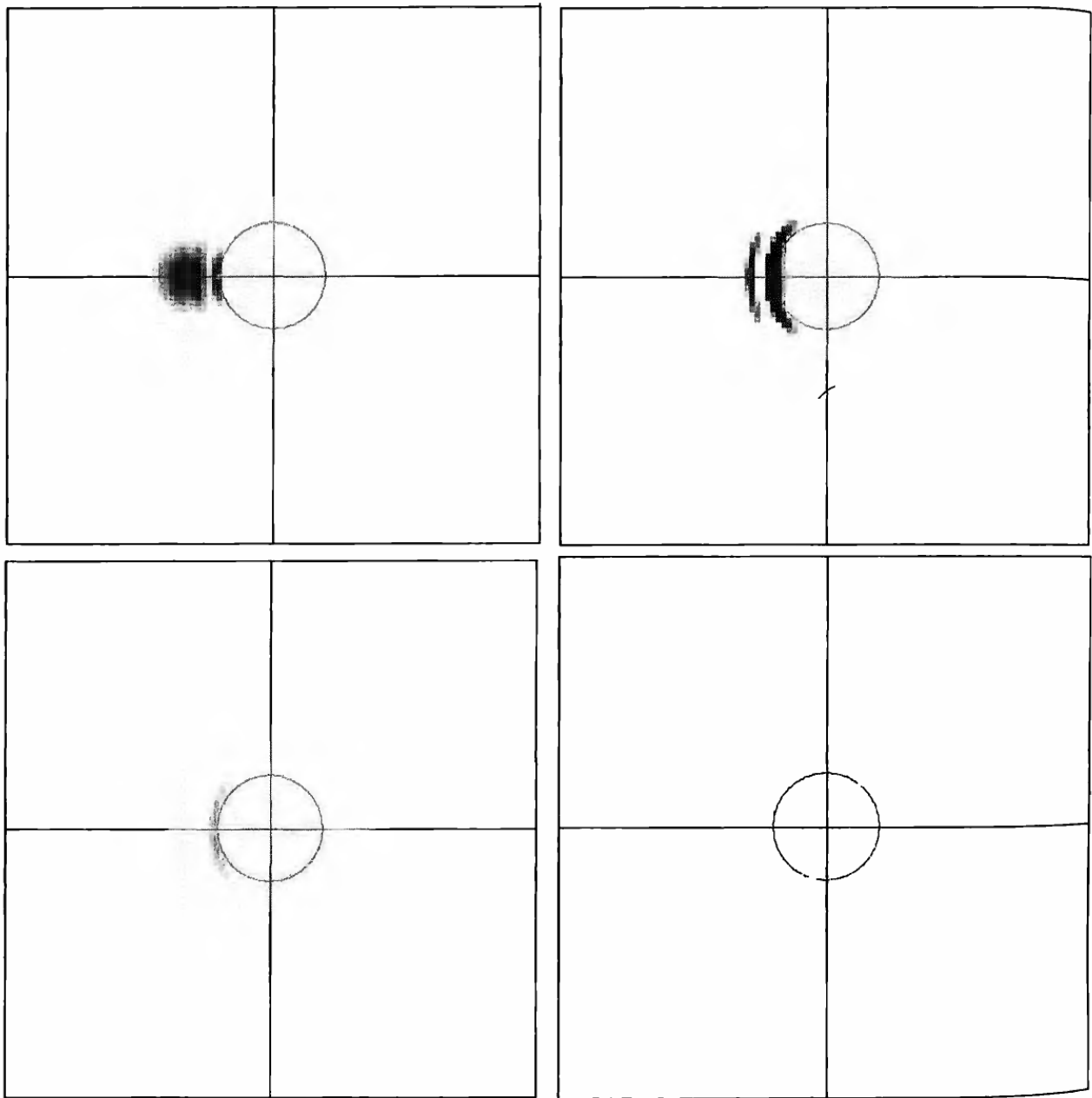


FIGURE 10.25: Wave packet with $k_0 = 40$, $\sigma = 0.1$, and $x_0 = -0.5$, impinging on a barrier of height $V_0 = 1000$. The shades of grey linearly progress toward black as the probability density rises. Shown at the upper left is the probability density at $t = 50\Delta t = 0.0050$, at $t = 0.0100$ (upper right), at $t = 0.0150$ (lower left), and $t = 0.0200$ (lower right). Space is discretized in units of $\Delta x = 0.02$, and the ranges of x and y are between -1 and 1 . The time step was $\Delta t = 0.0001$.

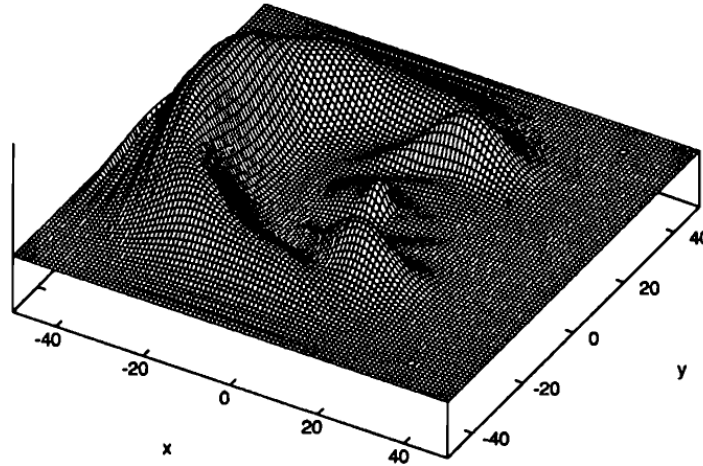


FIGURE 10.26: Wave packet of Figure 10.25 at $t = 250$ is shown in a three-dimensional mesh surface view. Note the small peak near the center of the potential barrier (the origin).

EXERCISES

- 10.19.** Write a program that implements the leapfrog algorithm in two dimensions, and use it to study two-dimensional scattering from cylindrical wells and barriers. Potential wells with a depth of twice the incoming kinetic energy, and barriers with a height of half the kinetic energy are interesting choices. Compare your results with those for the corresponding one-dimensional problem.

10.7 SPECTRAL METHODS

The time-dependent Schrödinger equation can also be treated with spectral methods, much like the approach we developed in our dealings with waves on a string. These methods make use of the formal time dependence of the wave function in (10.42), together with the fact that *any* initial wave function can be written in terms of the eigenstates of the associated time-independent problem. That is, if the functions φ_n are the solutions of the time-independent Schrödinger equation for the particular potential of interest, any wave packet ψ can be written as

$$\psi = \sum_n a_n \varphi_n, \quad (10.57)$$

where the a_n are complex constants. Note that here the label n is used to refer to different eigenstates, that is, energy levels; it does *not* correspond to space or time. This is analogous to the Fourier expansions of arbitrary functions, which are

discussed in Chapter 6 and Appendix C. The eigenfunctions φ_n form a complete set in the sense that *any* function can be written in the form (10.57).

The time dependence of the basis functions follows from (10.42). Since φ_n is an eigenfunction, we know that $\mathcal{H}\varphi_n = E_n\varphi_n$, where E_n is the energy of state n . If our initial wave function happened to be $\psi(t=0) = \varphi_n$, it would evolve in time according to (10.42), which leads to

$$\psi(t) = \exp(-itE_n/\hbar) \varphi_n . \quad (10.58)$$

For a wave packet composed of a sum of basis states such as (10.57) the same arguments give

$$\psi(t) = \sum_n a_n \exp(-itE_n/\hbar) \varphi_n . \quad (10.59)$$

Thus, if we can construct the expansion of the initial wave function in terms of the basis states, that is, if we can determine the coefficients a_n in (10.57), we can use (10.59) to compute the wave function at all future times.

This approach is very much like the spectral method we employed in Chapter 6. There we wrote the displacement of a string in terms of Fourier components. Since the time dependence of the Fourier components for waves on a string has a very simple form (purely sinusoidal), this allowed us to compute the time dependence of a wave packet on a string. Here we take similar advantage of the simple form (10.58) for the time dependence of a basis function. Evaluating (10.59) involves only some arithmetic with complex numbers; there are no algebraic or differential equations to solve.

To illustrate this approach we consider a one-dimensional harmonic oscillator. The first step is to determine the basis functions, φ_n . In this case these are the solutions to the time-independent Schrödinger equation, (10.2), with $V = Kx^2/2$. We considered this problem numerically in the exercises in Section 10.2. The exact solution is, of course, also known (the wave functions can be expressed as the product of a Gaussian and a Hermite polynomial), as discussed in Griffiths (1995) and Schiff (1968). However, since we want to show how the spectral approach works in the general case, we will calculate the basis functions numerically, using the shooting method. In principle, if we want the expansion (10.57) to represent our wave packet exactly, we need an *infinite* number of basis functions in the expansion. Of course, this is not possible in a numerical treatment; in the calculations below we will use 32 basis functions and find that this is sufficient for the particular problems we'll be considering. We will discuss how to estimate the necessary number of basis states in the exercises.

With the basis functions in hand, the next step is to calculate the expansion coefficients in (10.57). To do this we take advantage of the fact that the functions φ_n form an orthogonal set. That is, $\int \varphi_m^* \varphi_n dx$ is zero if $m \neq n$ and unity if $m = n$. Hence we can write

$$\int \varphi_m^* \psi_i dx = \int \left[\varphi_m^* \left(\sum_n a_n \varphi_n \right) \right] dx = a_m , \quad (10.60)$$

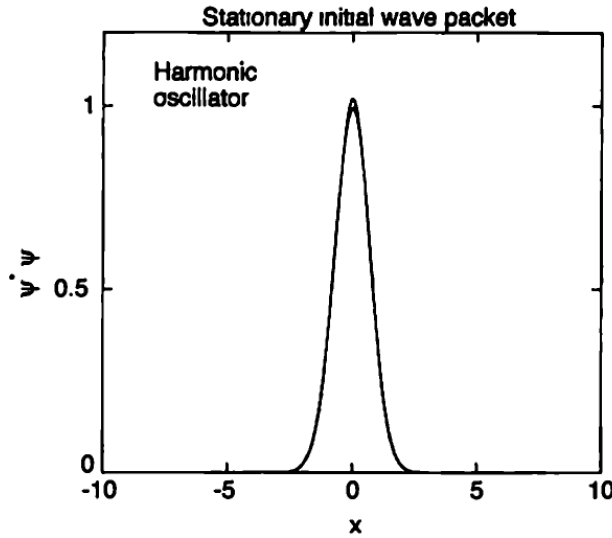


FIGURE 10.27: Time evolution of a stationary wave packet with an initial width $\sigma = 1$ [see (10.54)] The potential was $V = Kx^2/2$ with $K = 1$. We have plotted the wave packet at $t = 0, 0.5, 1$, and 2 , but the results overlap almost completely and are thus hard to distinguish

where ψ_i is our initial wave packet. Calculation of the expansion coefficients thus requires integration of the product of each basis function with the initial wave function. Since these functions are known only at the grid locations, this integration is actually just a simple sum.

The results of this calculation for a Gaussian initial wave packet (10.54) are shown in Figure 10.27. That is, we first used (10.60) to calculate the coefficients a_k , and then computed the variation of the wave function with time using (10.59). We see from Figure 10.27, that this wave packet changed very little with time. How can this be? It just so happens that we have picked an initial wave packet that is extremely close to the ground-state wave function, of the harmonic oscillator. The ground-state solution corresponds classically to a particle sitting at the bottom of the potential well ($x = 0$), and this is what we find in Figure 10.27.

An example that exhibits more “action” is shown in Figure 10.28, where we consider a narrower initial wave packet. The functional form of ψ_i is again (10.54) but here we have chosen $\sigma = 0.5$. This is considerably narrower than the ground-state wave function, so we are now far from being in an eigenstate, and the result is a substantial time dependence. Our packet was initially centered at $x = 0$, and it remained so as time advanced. However, the width grew considerably from $t = 0$ to 1.5 , providing another example of the uncertainty principle. The initial packet effectively contained components with wave vectors different from zero, and these components oscillated (harmonically) in the potential well. The initial symmetry was maintained, so the wave packet was always centered at $x = 0$, but the width varied with time. In fact, the width oscillated with time, as can be seen from the results plotted on the right in Figure 10.28.

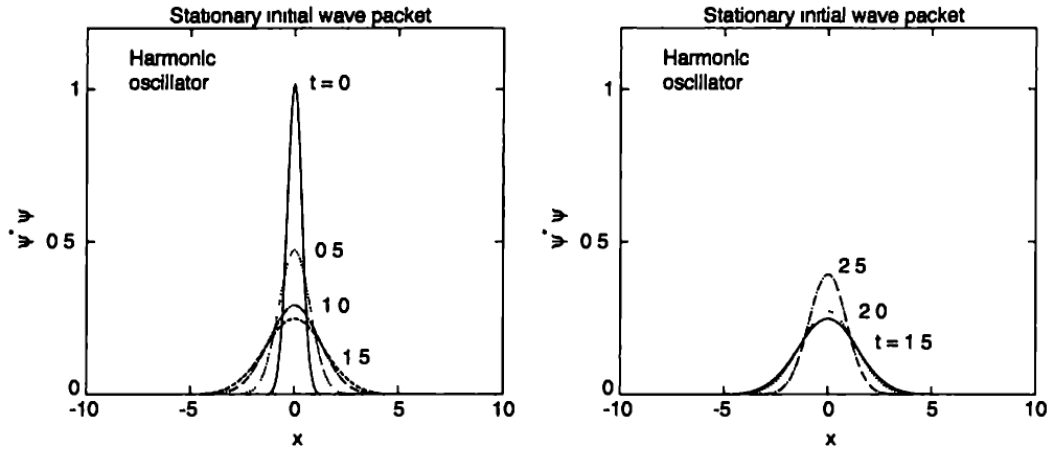


FIGURE 10.28: Time evolution of a wave packet with an initial width $\sigma = 0.5$. Left: $t = 0, 0.5, 1.0$, and 1.5 . Right: $t = 1.5, 2.0$, and 2.5 . The potential was the same as in Figure 10.27. Note that the packet initially broadened, but from $t = 1.5$ to 2.5 it became narrower with time.

It is also interesting to consider the effect of displacing the initial wave packet from $x = 0$, and this is shown in Figure 10.29. The initial wave packet was again given by (10.54), but here we took $x_0 = 2$ so that our packet was displaced from the center of the potential well. Classically this corresponds to giving the oscillator an initial displacement. We see that our particle oscillated back and forth in the well, precisely as expected for a simple harmonic oscillator.

Here the shape and width of the packet change very little with time, as we have chosen the value of σ that we have already shown to give an essentially time-independent width. We can therefore follow the position of the particle by simply plotting the location of the maximum of $\psi^*\psi$ as a function of time, and this is shown on the right in Figure 10.29. We find a simple cosine-like oscillation, as expected for a simple harmonic oscillator. We will leave it to the exercises to show that the period of these oscillations has the expected value (it does).

We have illustrated the spectral approach with a problem for which the basis states could have been calculated analytically (although we chose to calculate them numerically). However, this method can be used for problems in which little headway can be made analytically. To demonstrate this we consider the behavior of an anharmonic oscillator. We take the potential to be

$$V(x) = \frac{k_1}{2} x^2 + \frac{k_2}{2} x^4. \quad (10.61)$$

For small x the potential is nearly harmonic, while for large x the quartic part dominates. We have already considered this problem in an earlier exercise, where we used (or suggested that the reader use) the shooting method to calculate the eigenfunctions. Given these basis functions we can use (10.60) to calculate the expansion coefficients for any desired initial wave packet. The time dependence of the basis functions, which follows from the energy eigenvalues, then leads to the time dependence of the complete wave packet.

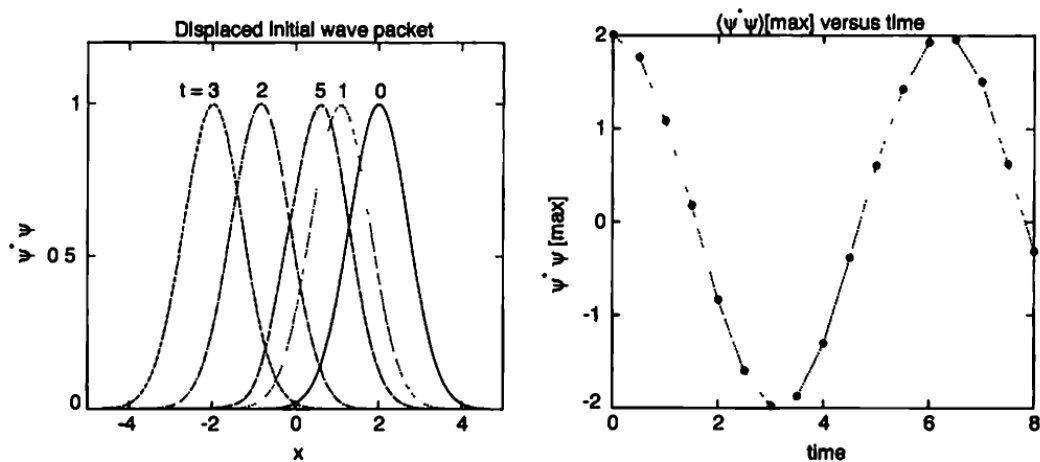


FIGURE 10.29: Left: time evolution of a wave packet with an initial width of $\sigma = 1$, and an initial displacement of $x_0 = 2$ [in (10.54)]. Right: location of the maximum in $\psi^* \psi$ as a function of time. From a "classical" viewpoint this corresponds to plotting the position of the particle as a function of time. The result is simple harmonic motion.

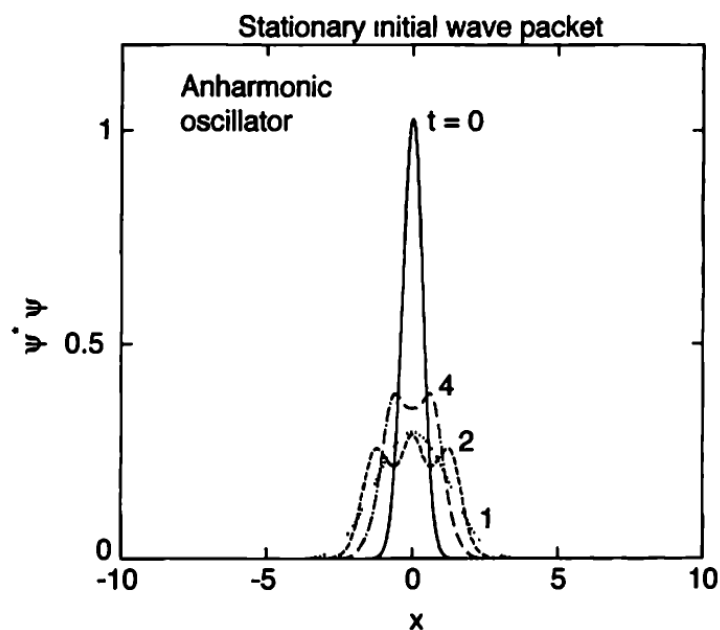


FIGURE 10.30: Time evolution of a wave packet in an anharmonic potential well, with an initial width $\sigma = 0.5$. The potential was given by (10.61) with $k_1 = 1$ and $k_2 = 0.1$. The lowest 32 basis states were used in the calculation.

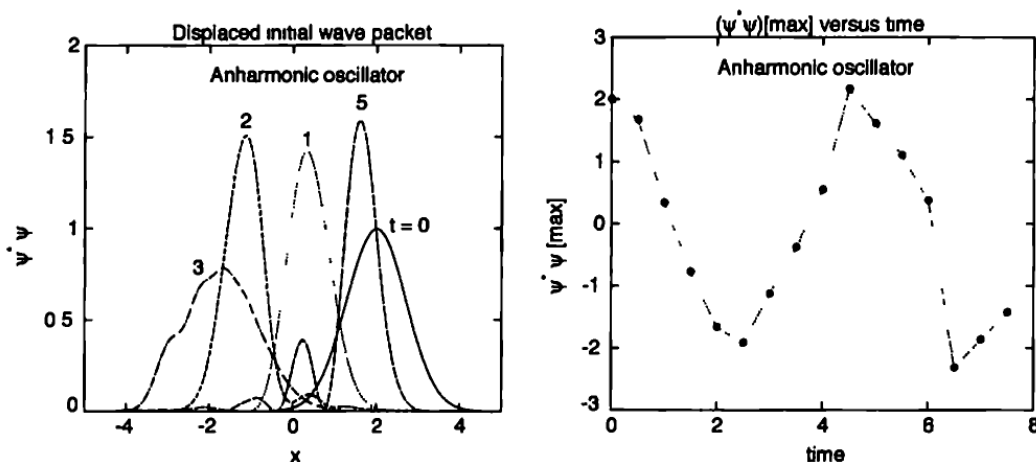


FIGURE 10.31: Time evolution of a wave packet with an initial width $\sigma = 1$ and an initial displacement of $x_0 = 2$, in an anharmonic potential well. The other parameters are the same as in Figure 10.30.

Results for a stationary initial wave packet, with $\sigma = 0.5$, are shown in Figure 10.30. This is the same wave packet considered for the harmonic oscillator potential in Figure 10.28. However, the results are very different due to the change in the potential, as we now consider in more detail. While the wave packet remained centered at $x = 0$, its width oscillated with time in a manner reminiscent of the purely harmonic case. However, here the *shape* of the wave packet also changed with time. This was due to the anharmonicity of the potential. For a purely harmonic system the period of oscillation is independent of the amplitude, but this is not true for an anharmonic oscillator. For a potential proportional to x^4 the period *decreases* as the amplitude is increased (see Figure 10.13). This can be understood in terms of the increasing steepness of the potential at large x . As compared with the harmonic case, this confines the particle to a smaller region of space, thereby decreasing the period.

This dependence of the period on amplitude affects the wave packet in the following way. We have already noted that our Gaussian packet, or indeed any other packet, can be thought of as a sum of basis functions with different effective wave vectors; that is, with different velocities, with some of the components moving to the right and others to the left. For the harmonic case the components oscillated with precisely the *same* period, and this preserved the shape of the initial wave packet. However, for the anharmonic case the components have different periods, and this causes them to get out of step with each other as time passes. One result is a distortion of the shape of the wave packet.

This anharmonicity can also be seen in the behavior of a wave packet that is initially displaced from $x = 0$. An example is shown in Figure 10.31 where we plot the time evolution of a packet that was initially centered at $x = 2$. The particle oscillated back and forth in the well but the shape of the packet varied considerably, in contrast to the behavior in the purely harmonic case (Figure 10.29). The time dependence of the location of the maximum in $\psi^*\psi$ also deviated substantially from a purely cosine form.

The relatively simple behavior of a harmonic oscillator is not, in general, found for other potentials. The contrasting results for an anharmonic oscillator illustrate some of these features. These examples also highlight the power of the spectral method and show especially that it can be applied to problems for which analytic results are not possible.

EXERCISES

- 10.20. Consider the harmonic oscillator and calculate the evolution of a wave packet that is initially at $x = 0$ and has a wave vector $k = 0.5$ [see (10.55)]. Show that it oscillates, and compare its period with that found when the wave packet is given an initial displacement, but no initial velocity.
- *10.21. Perform a spectral calculation for a harmonic oscillator, and investigate how the results depend on the number of basis states that are used. Contrast the results with 8 and 16 basis functions with the results shown for 32 basis states in Figure 10.29. As a general rule, the basis functions vary more rapidly with position as their energy is increased. These functions must vary at least as rapidly as the initial wave function (if not more so), if they are to be able to accurately represent the time dependence. To see this, perform a calculation with a fixed number of basis states and compare the results for different initial displacements of the wave packet (as in Figure 10.29).
- 10.22. Calculate the (classical) oscillation period for the harmonic oscillator considered in Figure 10.29 and show that it agrees with the observed behavior.

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